

April 30, 2026



Reinforcement Learning-based Fine-Tuning (ReFT) for Deep Video Understanding

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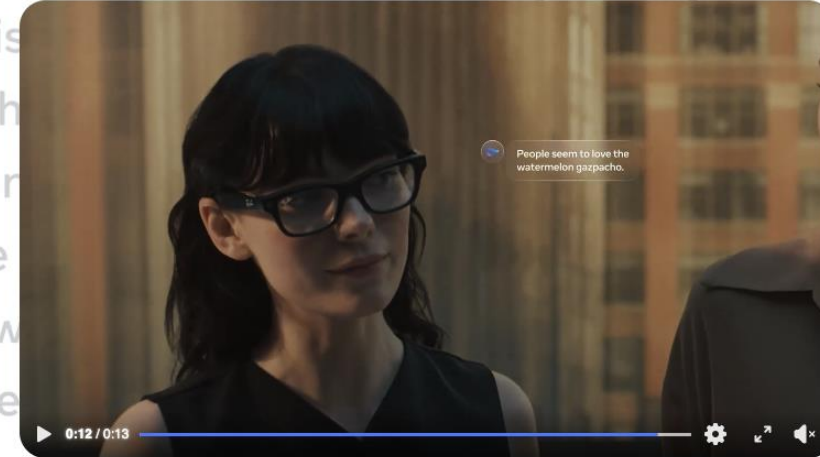


Multimodal Streaming Videos from Smart Glasses



<https://ego-exo4d-data.org/>

translate this sign, recommend a side dish for salmon,
remind me to leave work early for my son's play,
identify this...
tell me how to make a French,
the recipe for Grandma,
the oven,
to leave,
of this hotel,
suggest a good activity that will be fun for kids,
and to pick up Mom and Dad at the airport tonight



<https://www.meta.com/kr/en/ai-glasses/ray-ban-meta/>

How can we evaluate his video understanding?

Start with simple task. Video classification!



Apply Eye Make
Playing Guitar
Baby Crawling

(UCF101, classes)

?



Apply Eye Makeup



Playing Guitar



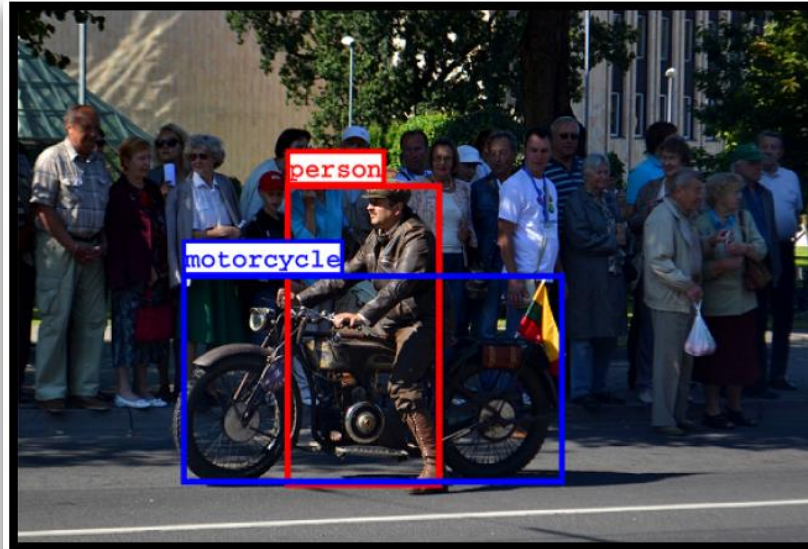
Baby Crawling

Harder task? Human-object interaction detection

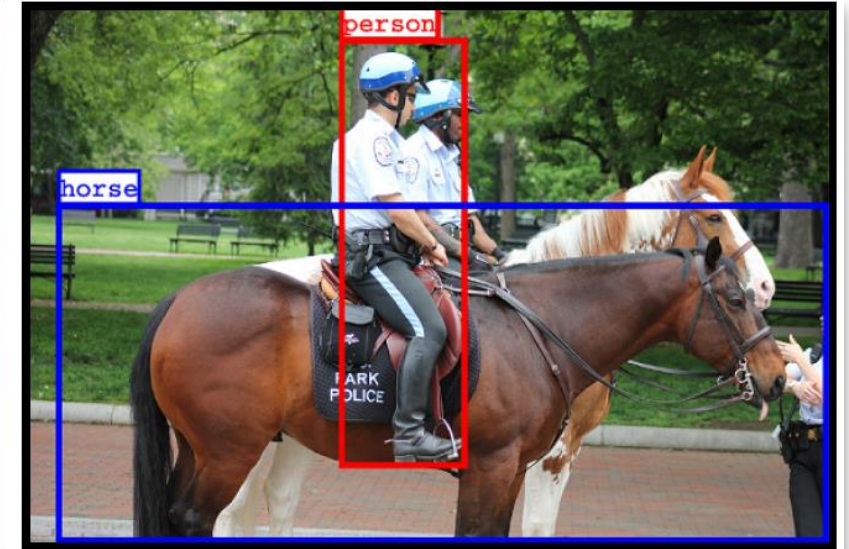
Detecting **objects** and classify their **interactions**.



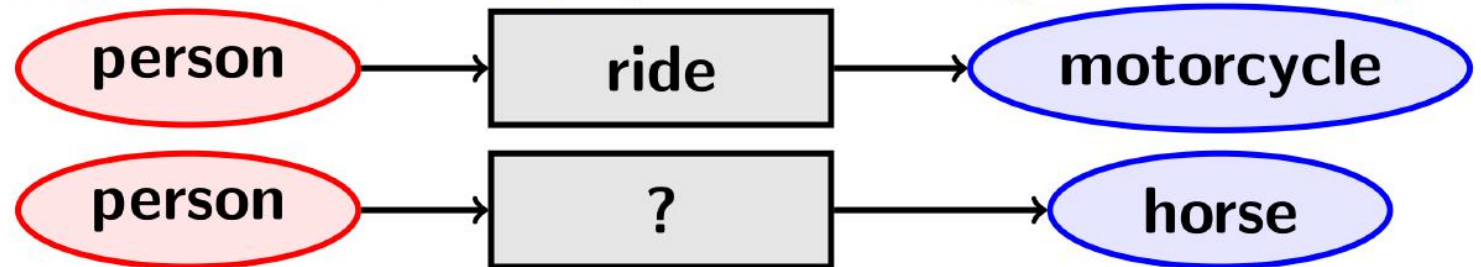
?



(a) $[[\text{person}, \text{ride}, \text{motorcycle}]]$



(b) $[[\text{person}, ?, \text{horse}]]$



(c) Objects interactions with predicates

- **Task definition:**
 1. Object detection
 2. Interaction Classification

- **Goal:**
< **Human**, Interaction , **Obj** >

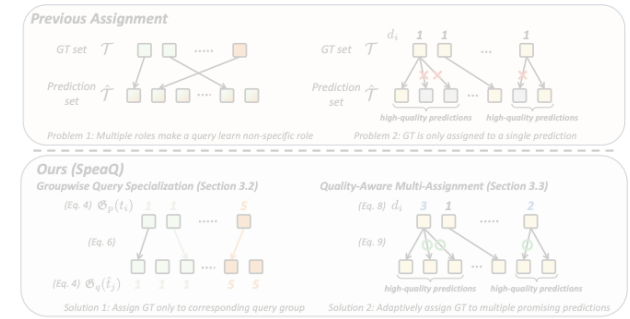
High-level Scene Understanding

Beyond pre-defined classes,
can we ask hard free-form questions?

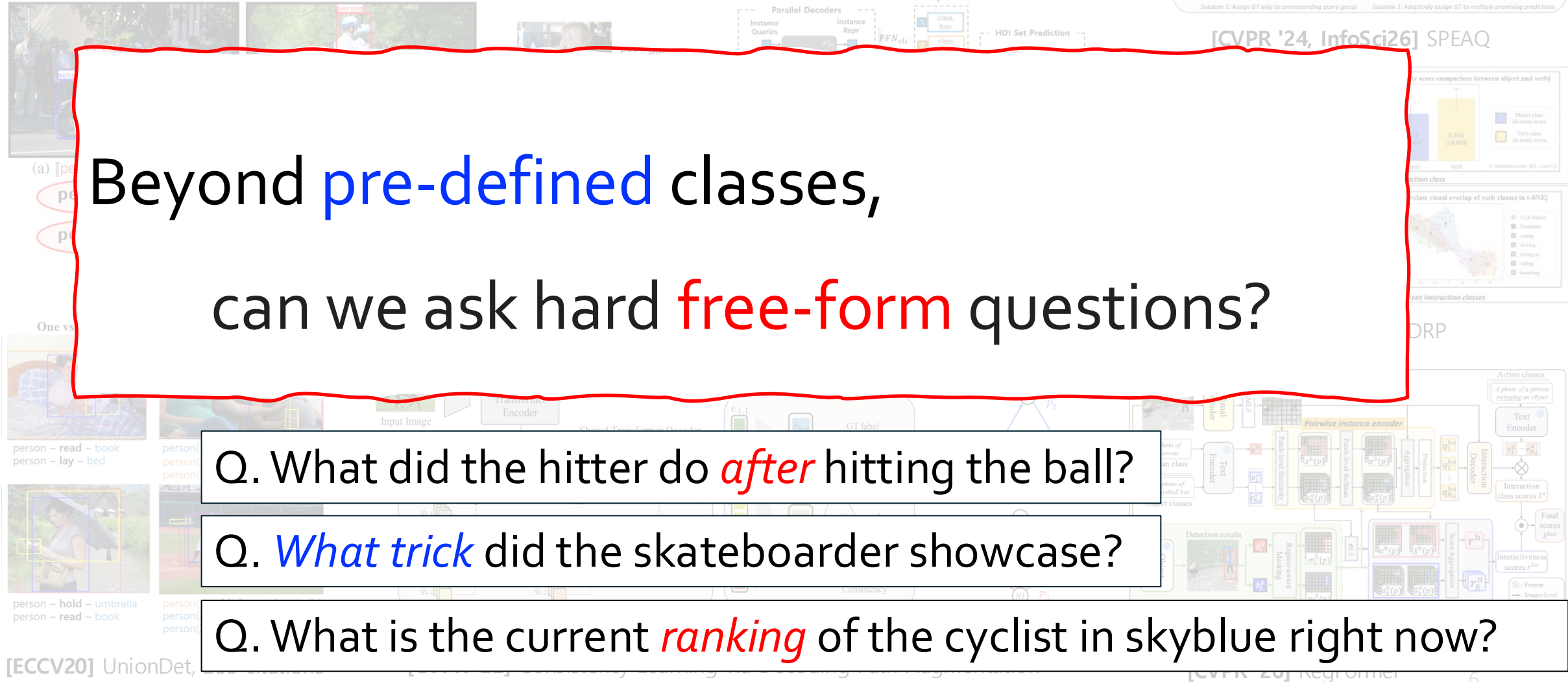
Q. What did the hitter do *after* hitting the ball?

Q. *What trick* did the skateboarder showcase?

Q. What is the current *ranking* of the cyclist in skyblue right now?



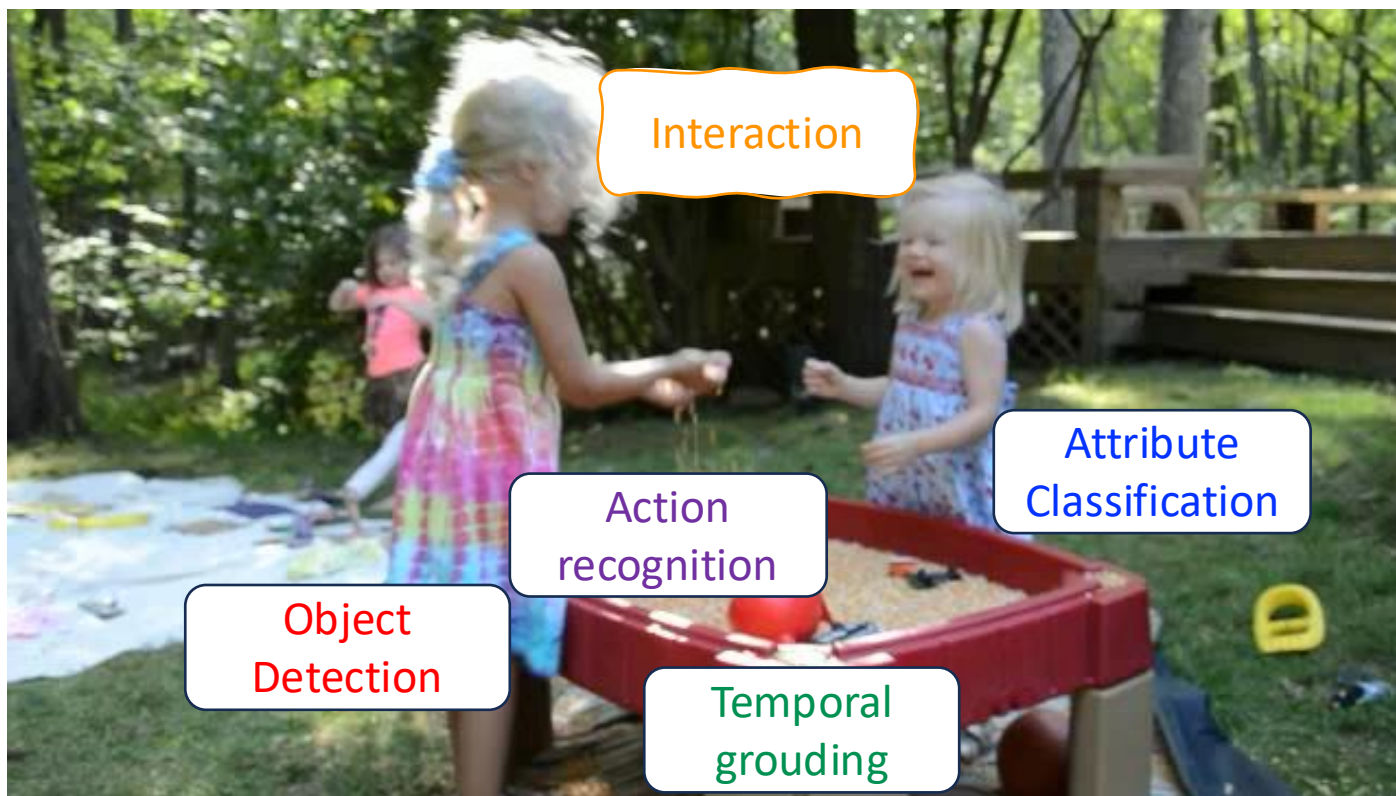
[ICVPR '24, InfoSci26] SPEAQ



Video Question Answering

< Video, Question, Answer >

Video:



Question:

How did the girl in blue respond each time the girl in multicoloured dress picks up some sand?

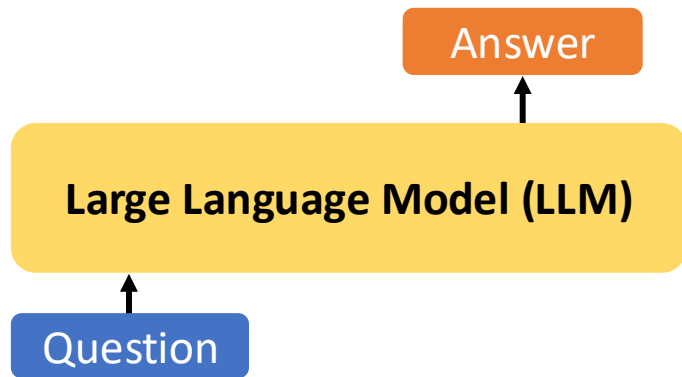
Choices:

- (A) Watches the girl
- (B) Sit down
- (C) Knock away the sand
- (D) Pull her away
- (E) Grab the sand

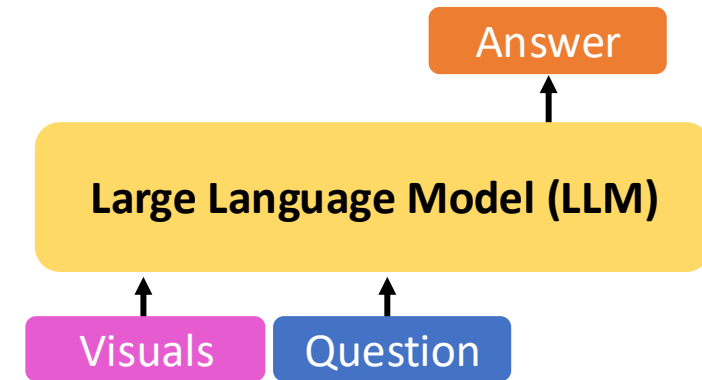
Answer: (C)

Question Answering with LLM

QA (text only)



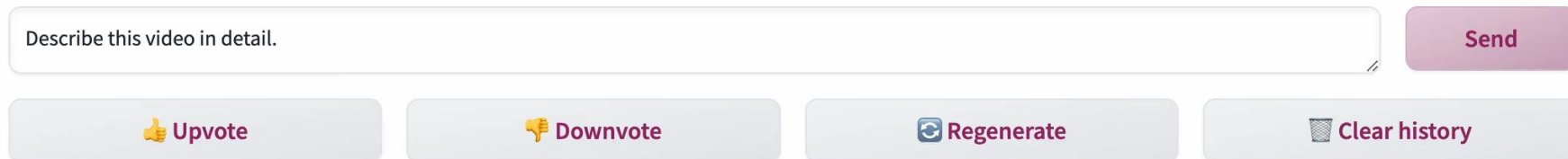
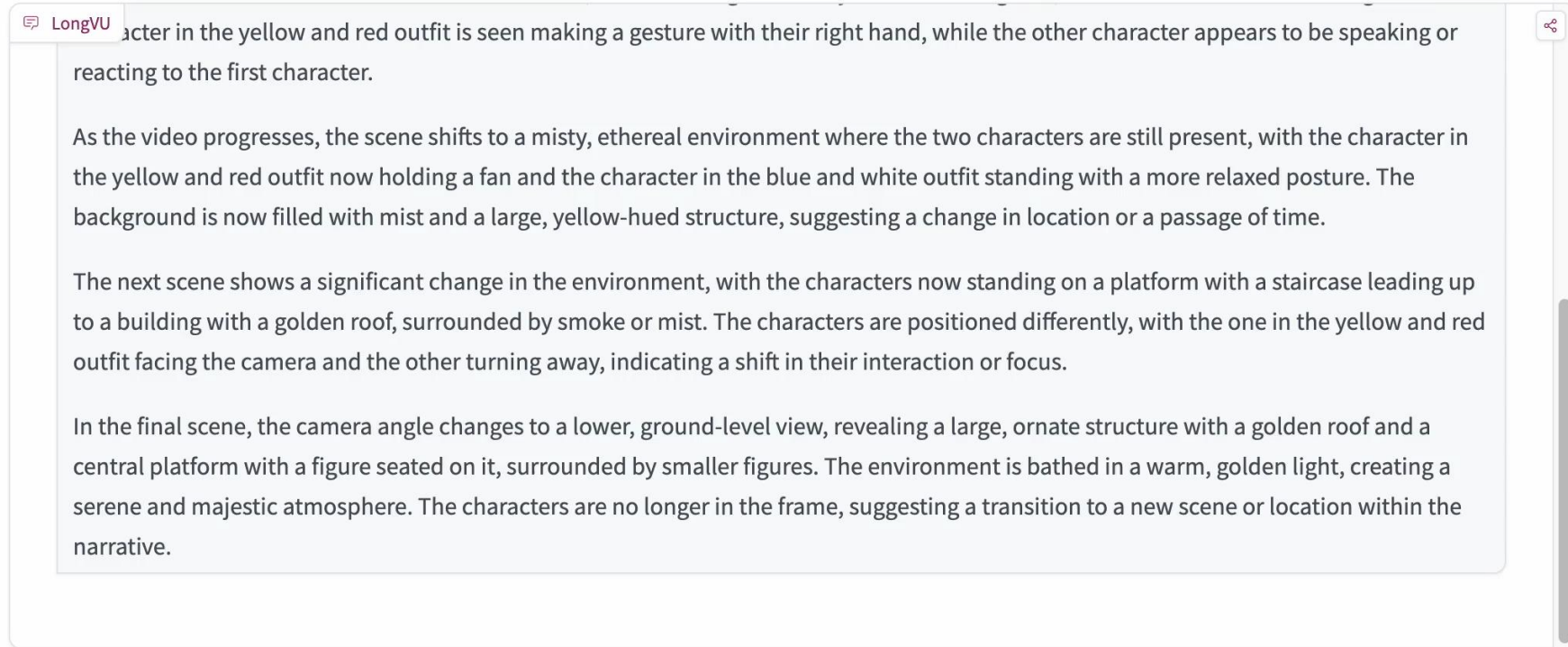
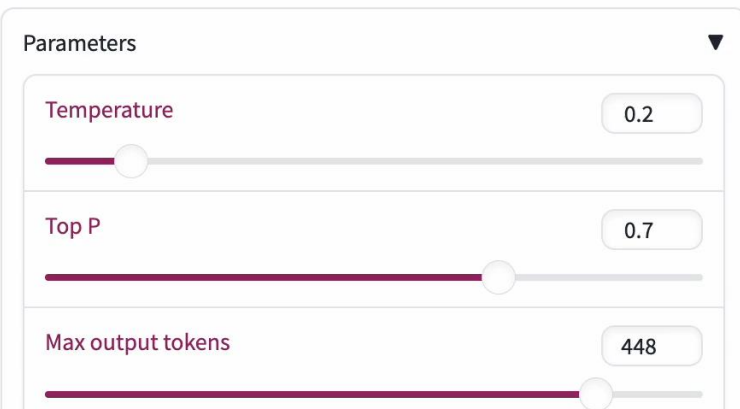
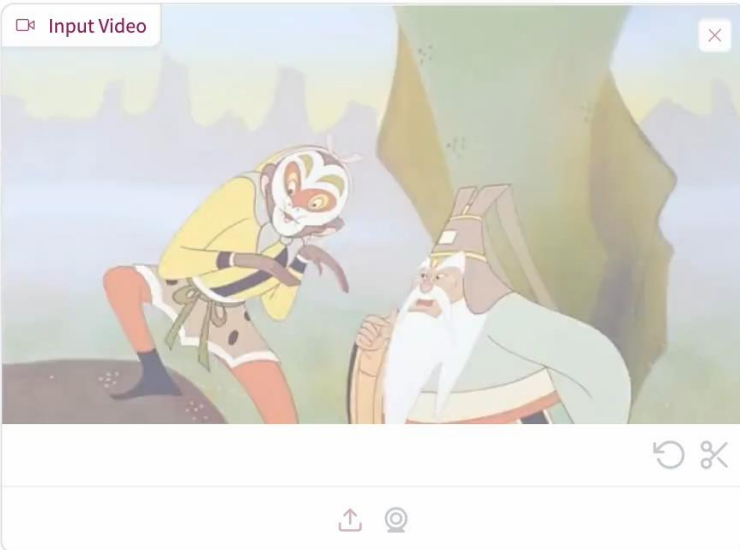
Video QA (video+text)



Adaptation from text (unimodal) to video+text (multimodal)

Interactive Video Understanding/MLLM

LongVU



<https://huggingface.co/spaces/Vision-CAIR/LongVU>

Interactive Video Understanding/MLLM

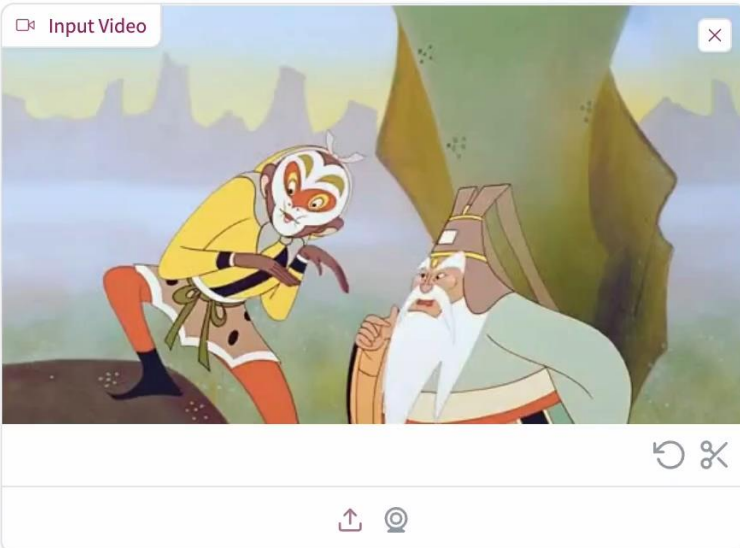
LongVU



KAUST



MLV
Machine Learning & Vision Lab



Parameters

Temperature

0.2

Top P

0.7

Max output tokens

448

LongVU

Character in the yellow and red outfit is seen making a gesture with their right hand, while the other character appears to be speaking or reacting to the first character.

As the video progresses, the scene shifts to a misty, ethereal environment where the two characters are still present, with the character in the yellow and red outfit now holding a fan and the character in the blue and white outfit standing with a more relaxed posture. The background is now filled with mist and a large, yellow-hued structure, suggesting a change in location or a passage of time.

The next scene shows a significant change in the environment, with the characters now standing on a platform with a staircase leading up to a building with a golden roof, surrounded by smoke or mist. The characters are positioned differently, with the one in the yellow and red outfit facing the camera and the other turning away, indicating a shift in their interaction or focus.

In the final scene, the camera angle changes to a lower, ground-level view, revealing a large, ornate structure with a golden roof and a central platform with a figure seated on it, surrounded by smaller figures. The environment is bathed in a warm, golden light, creating a serene and majestic atmosphere. The characters are no longer in the frame, suggesting a transition to a new scene or location within the narrative.

Describe this video in detail.

Send

Upvote

Downvote

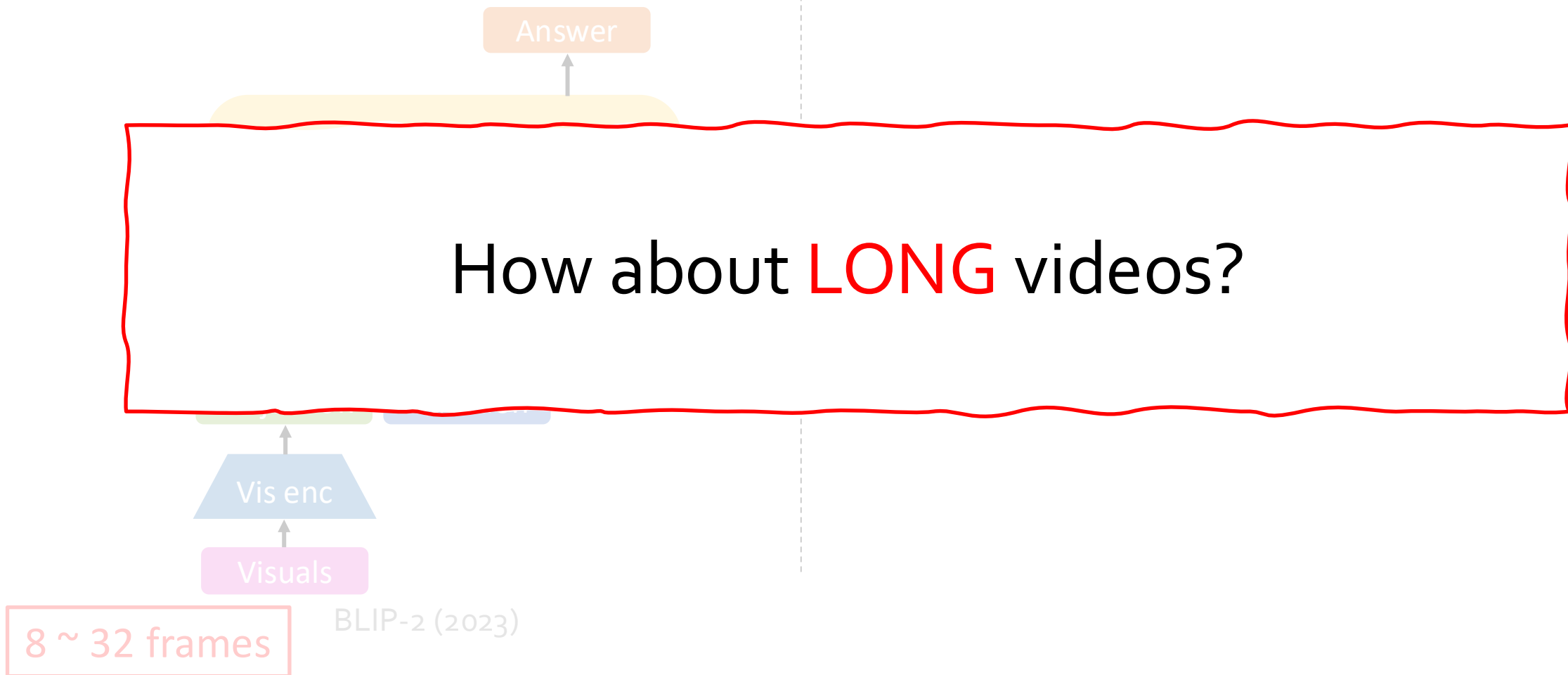
Regenerate

Clear history

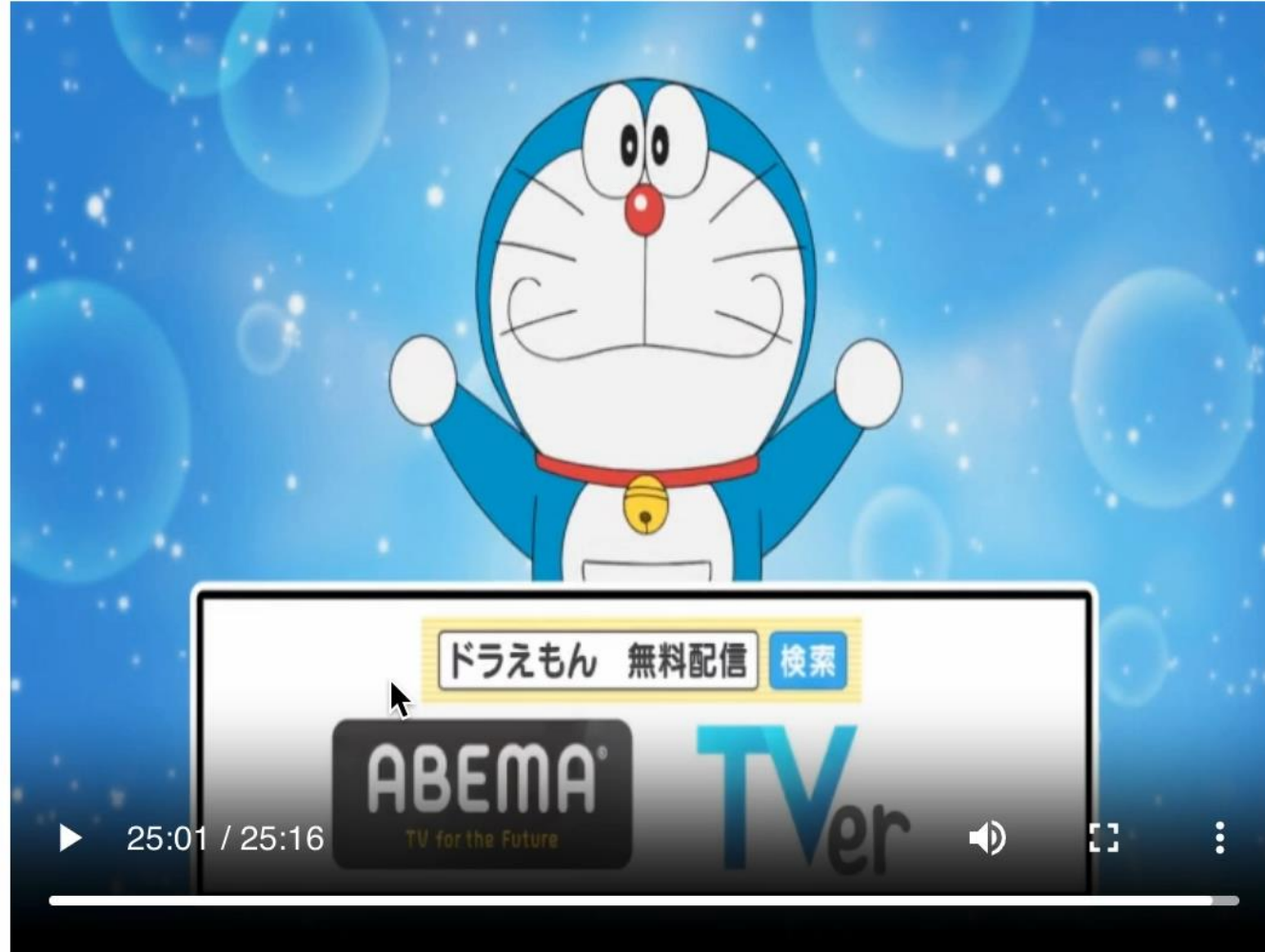
<https://huggingface.co/spaces/Vision-CAIR/LongVU>

MLLM: LLaMA-Adapter

LLM + visual inputs



Long Video Examples



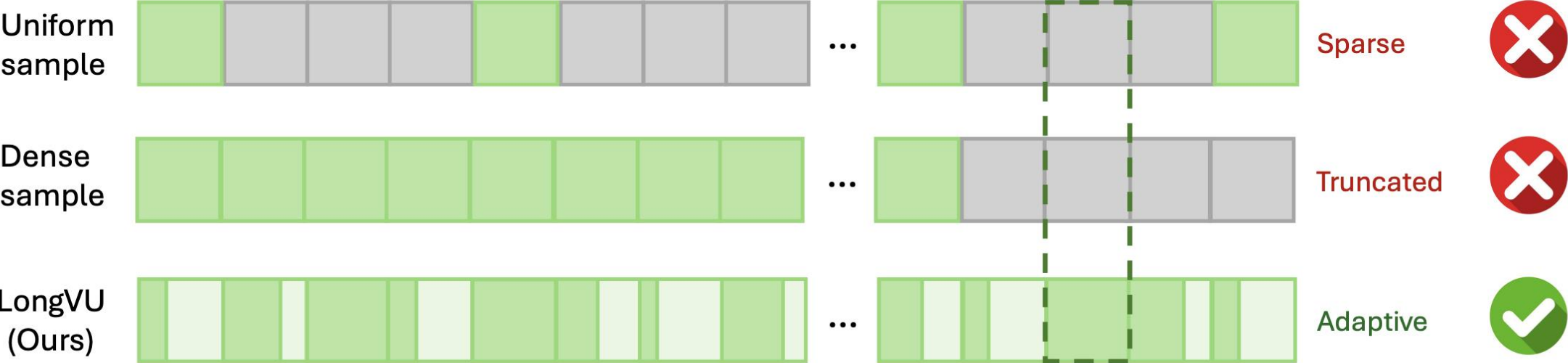
User: What is the condition of the highway where the SUV is parked?

Long Video Examples



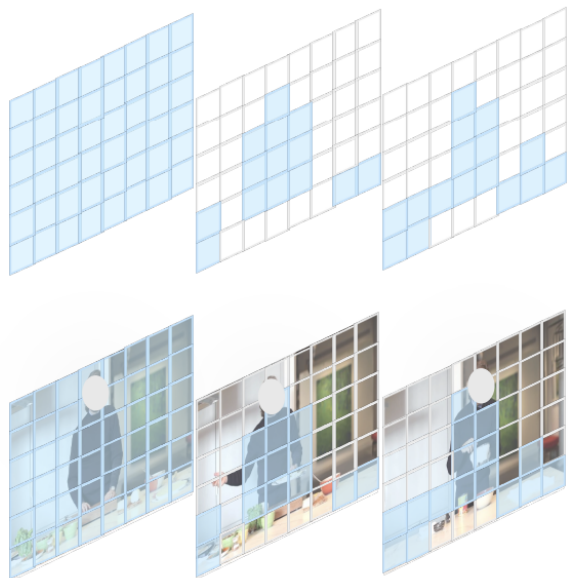
User: What is the condition of the highway where the SUV is parked?

Question: Where did I put the white cup?



You left the white cup on the table.

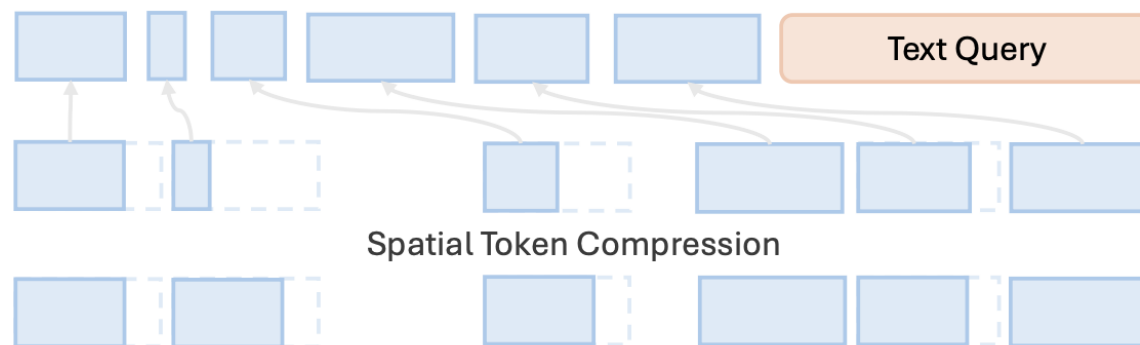
Spatial Token Compression



Temporal Reduction

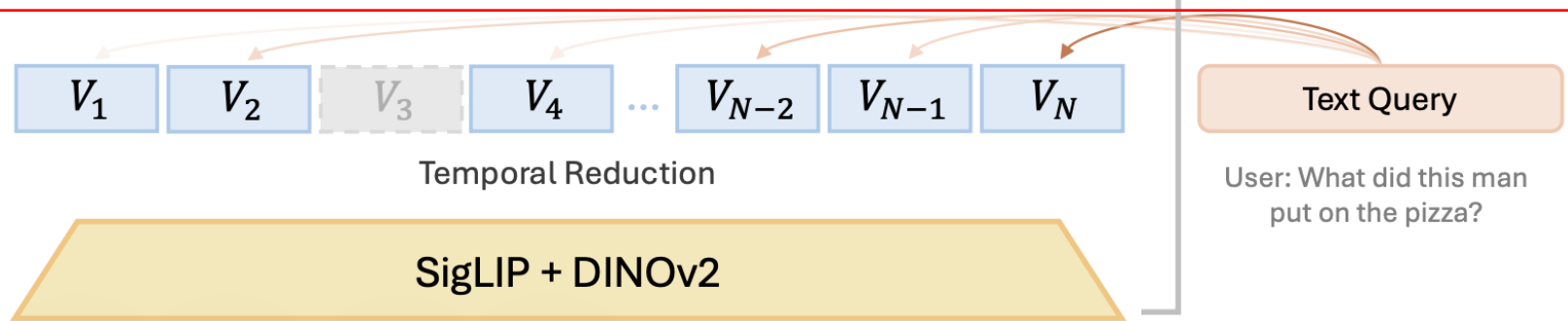


Large Language Model



Spatial Token Compression

Spatiotemporal Adaptive Token Compression

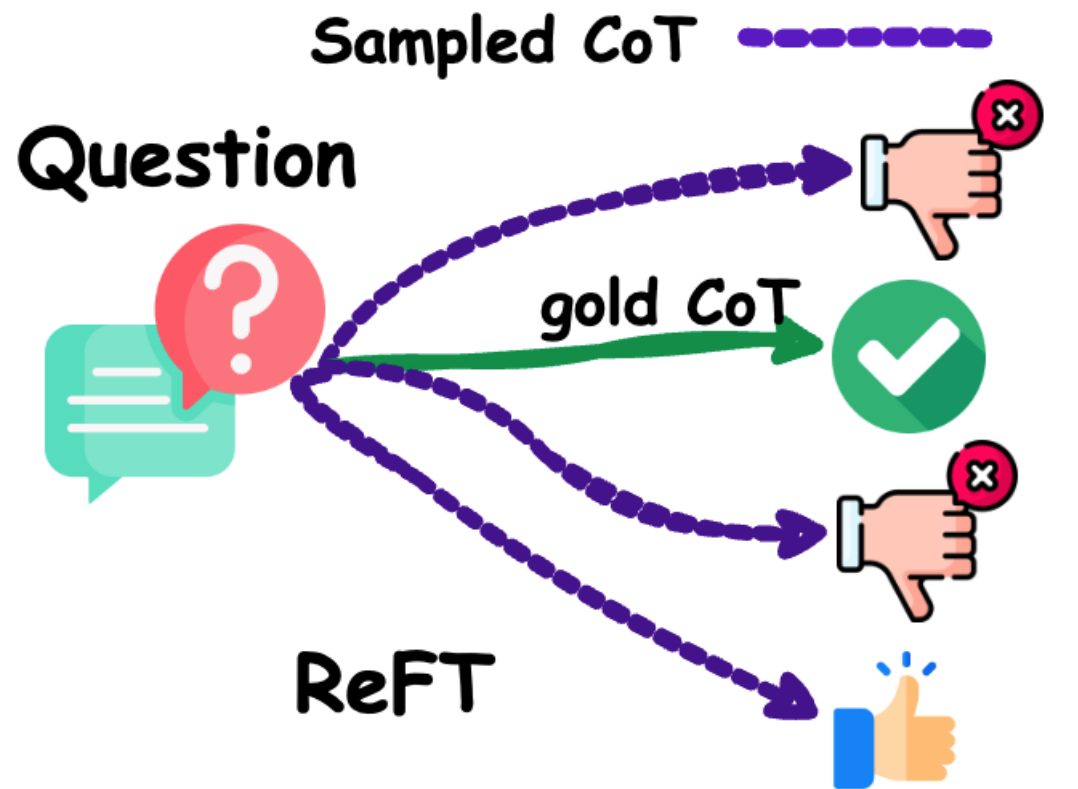
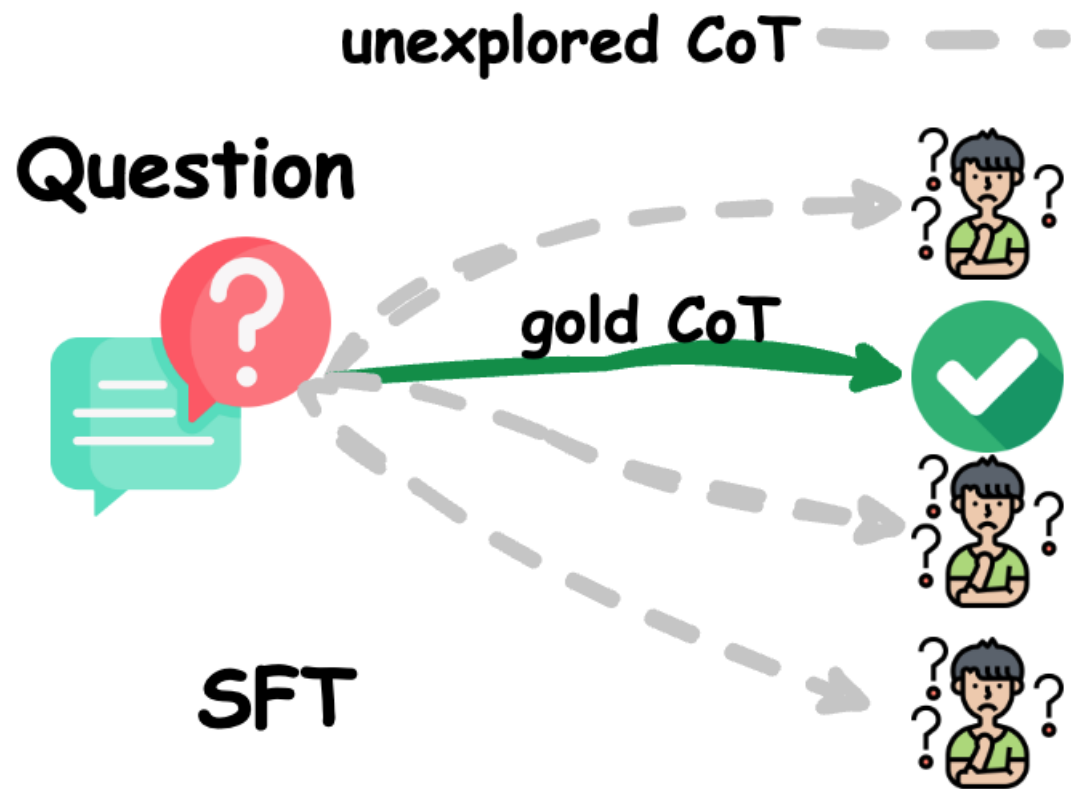


Temporal Reduction

SigLIP + DINOv2



Reinforcement Fine-Tuning



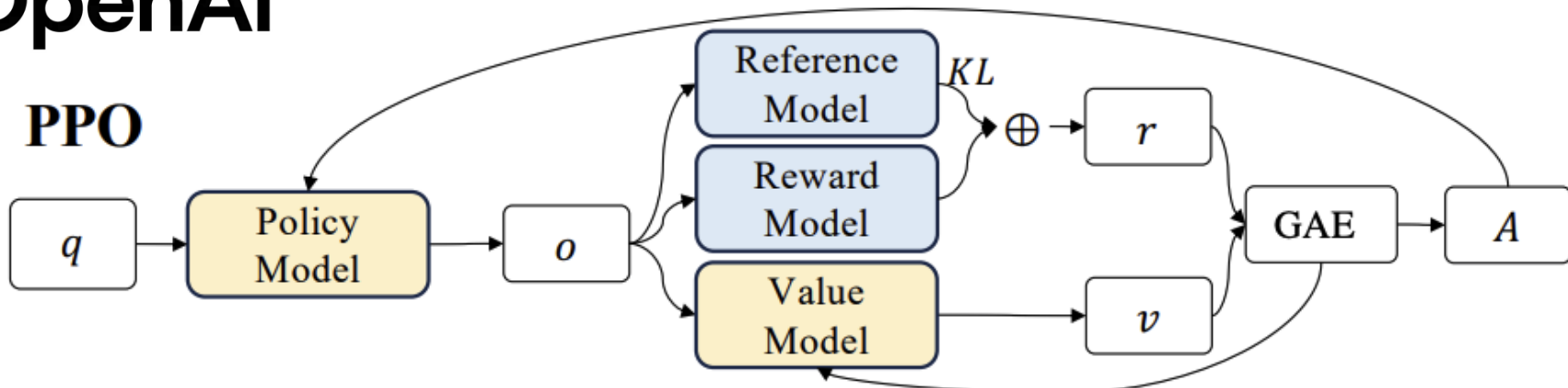
Reward/Feedback

Preliminary

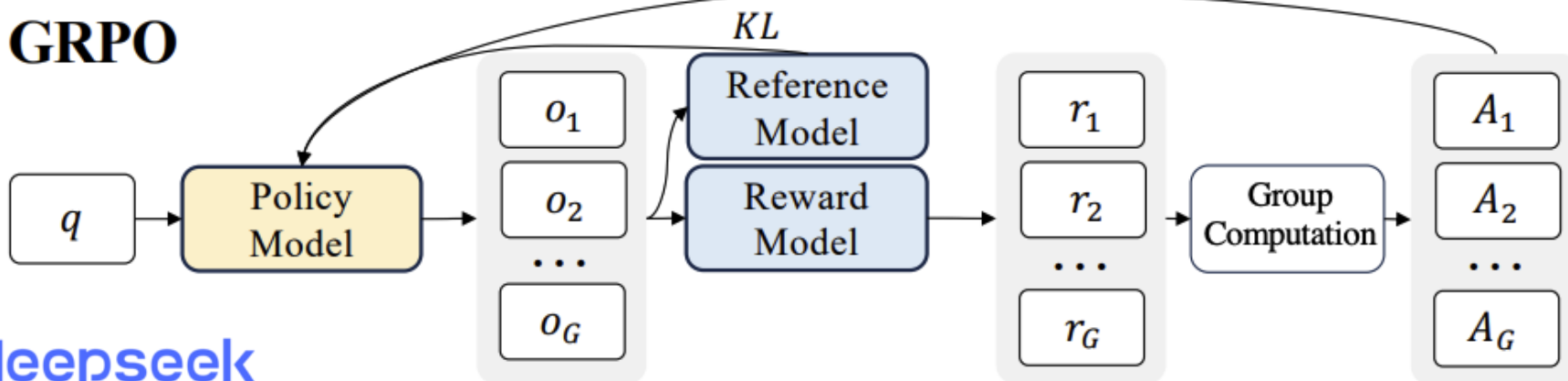


Proximal Policy Optimization

PPO



GRPO



Trained
Models

Frozen
Models



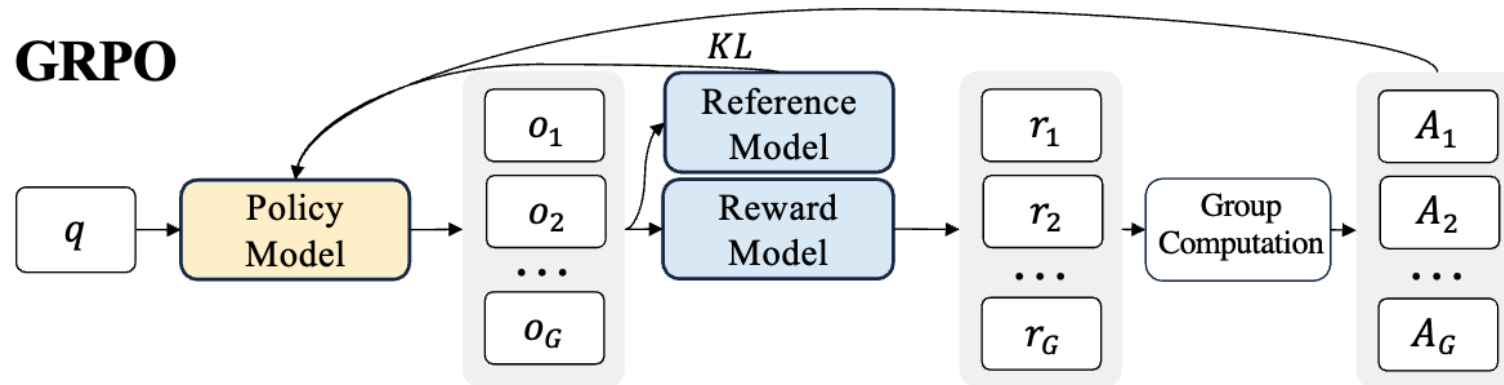
$$J_{\text{PPO}}(\theta) = \mathbb{E}_{q \sim P(Q), o \sim \pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} \min \left(\frac{\pi_{\theta}(o_t | q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t | q, o_{<t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t | q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t | q, o_{<t})}, 1 - \epsilon, 1 + \epsilon \right) A_t \right) \right]$$

Preliminary

Group Relative Preference Optimization

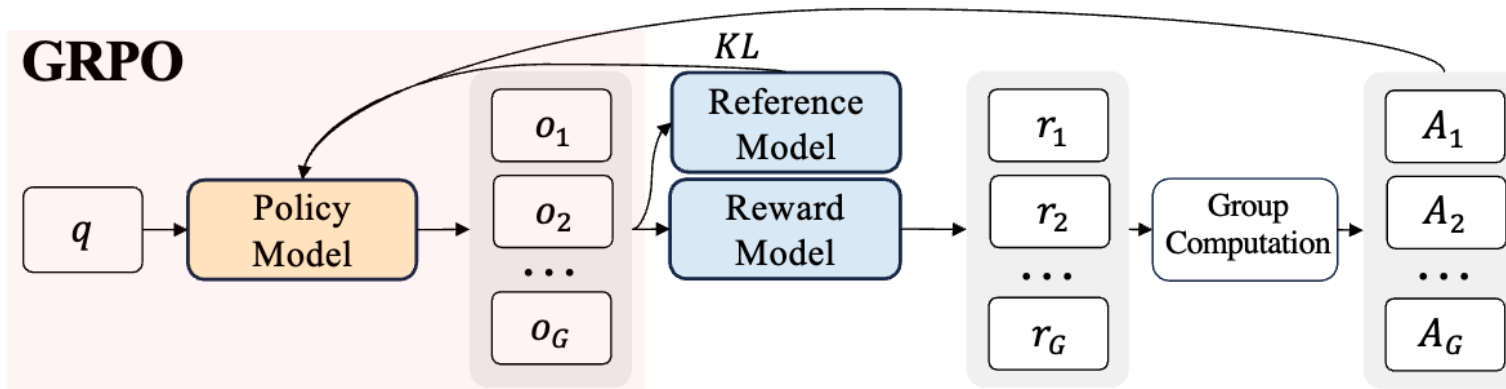


GRPO



Preliminary

Group Relative Preference Optimization



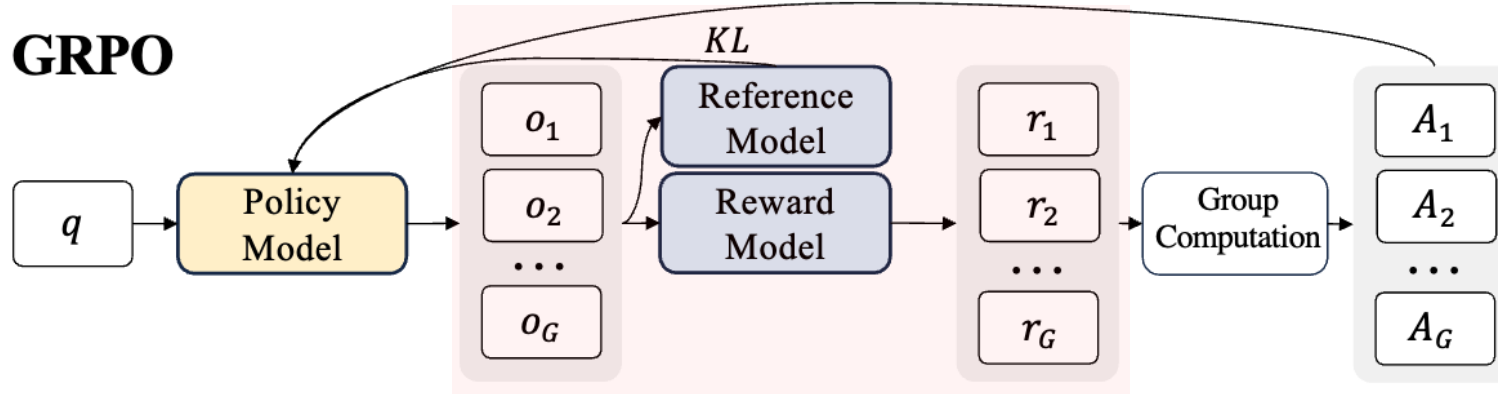
1. Generate multiple responses

Sample multiple responses from the policy for the same input

Preliminary

Group Relative Preference Optimization

GRPO

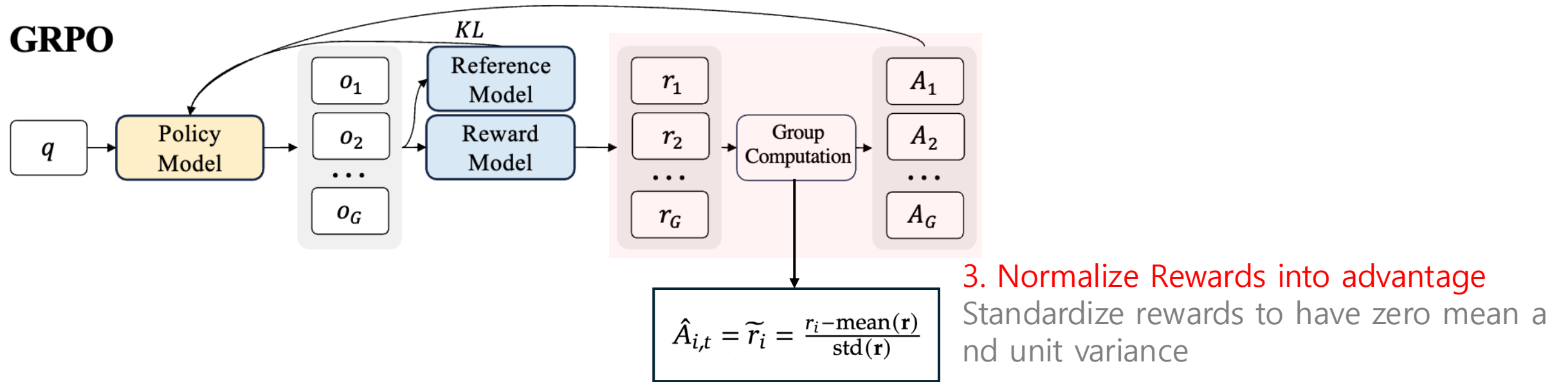


2. Evaluate each response

Use a reward model or rule-based reward functions (e.g. format, task-specific metrics)

Preliminary

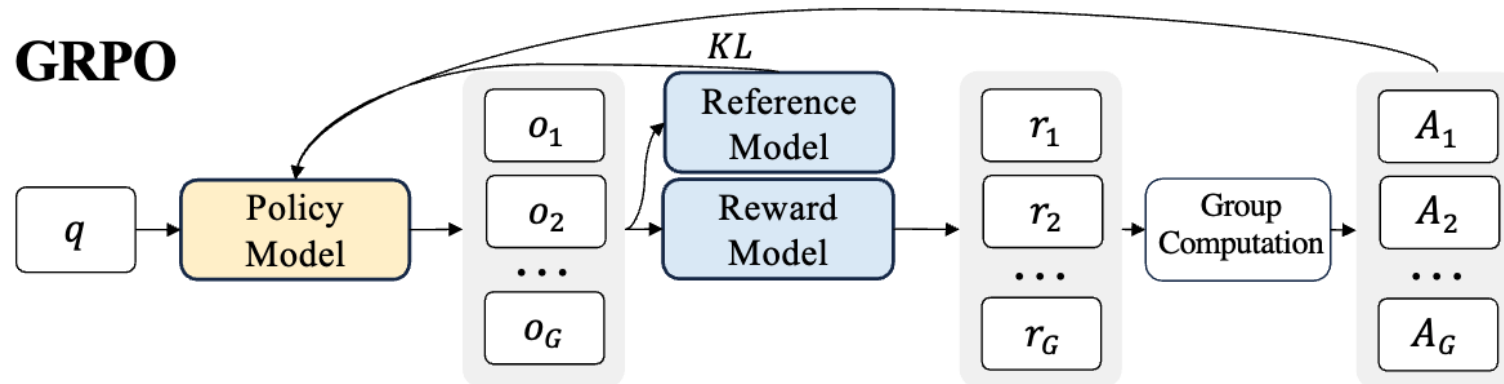
Group Relative Preference Optimization



Preliminary

Group Relative Preference Optimization

GRPO



4. Optimize the model by maximizing the objective function below

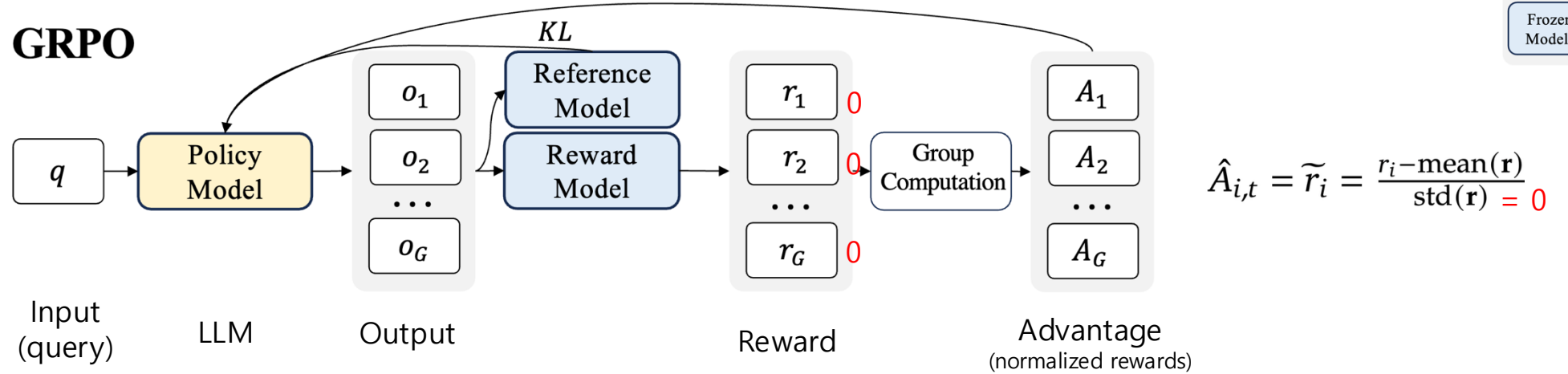
Update the policy model π_θ to increase the probability of high reward responses

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} [\pi_\theta || \pi_{ref}] \right\}$$

Limitations of GRPO (on-going project/under review)

GRPO



$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} [\pi_{\theta} || \pi_{ref}] \right\} \right]$$

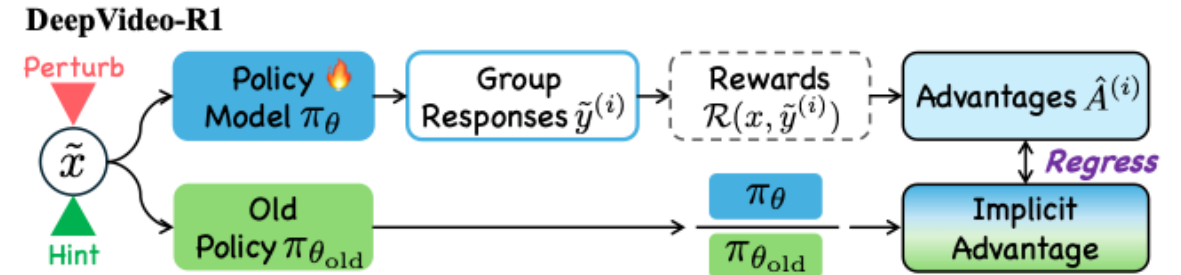
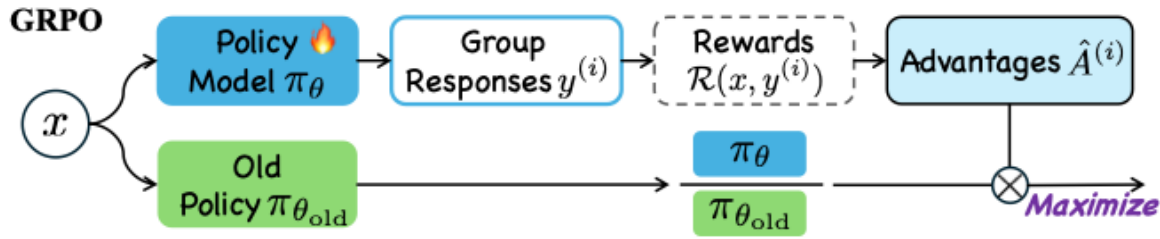
$$q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q).$$

Problem1. Safeguards are needed for stable learning

Problem2. Fails to obtain effective learning signals from similar responses (**too hard** or too easy samples)

Problem1. Safeguards are needed for stable learning

Sol 1. **Reg**-GRPO (Regressive GRPO)



- Reframes advantage maximization as advantage regression
- GRPO**: Maximizes the probability of good response

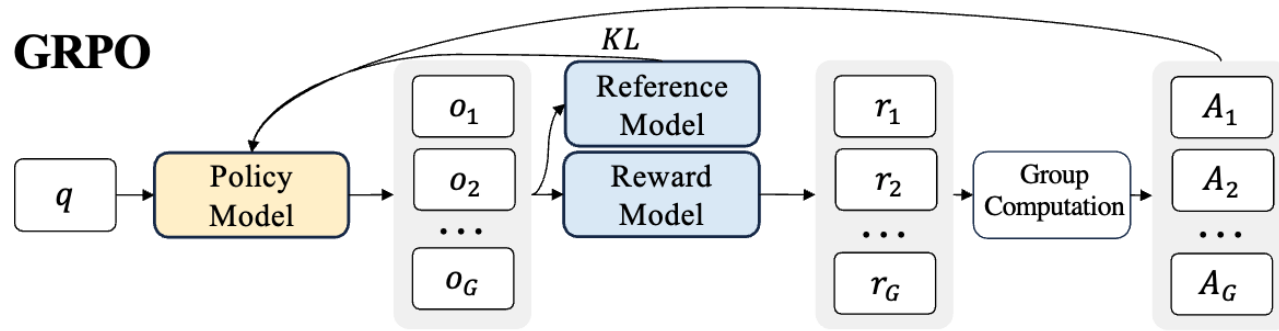
$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\mathbf{x} \sim P(X)}$$

$$- \frac{1}{G} \sum_{i=1}^G \sum_{t=1}^{|\mathbf{y}^{(i)}|} \min \left[\frac{\pi_\theta(y_t^{(i)} | \mathbf{x}, \mathbf{y}_{<t}^{(i)})}{\pi_{\theta_{old}}(y_t^{(i)} | \mathbf{x}, \mathbf{y}_{<t}^{(i)})} \hat{A}^{(i)}, \text{clip} \left(\frac{\pi_\theta(y_t^{(i)} | \mathbf{x}, \mathbf{y}_{<t}^{(i)})}{\pi_{\theta_{old}}(y_t^{(i)} | \mathbf{x}, \mathbf{y}_{<t}^{(i)})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}^{(i)} \right]$$

- Reg-GRPO**: Regresses to align the implicit reward with advantages derived from actual reward

$$\mathcal{L}_{\text{reg-grpo}} = \mathbb{E}_{\mathbf{x} \sim P(X)} \frac{1}{G} \sum_{i=1}^G \left(\hat{A}^{(i)} - \epsilon \cdot \frac{\log \rho(\mathbf{x}, \mathbf{y}^{(i)}) - \mu_\rho}{\sigma_\rho} \right)^2. \quad \hat{A}^{(i)} = \epsilon \cdot \frac{(\log \rho^*(\mathbf{x}, \mathbf{y}^{(i)}) - \mu_\rho^*)}{\sigma_\rho^*}, \quad \rho^*(\mathbf{x}, \mathbf{y}) = \frac{\pi_\theta^*(\mathbf{y} | \mathbf{x})}{\pi_{\theta_{old}}^*(\mathbf{y} | \mathbf{x})},$$

Problem2. Vanishing Advantage



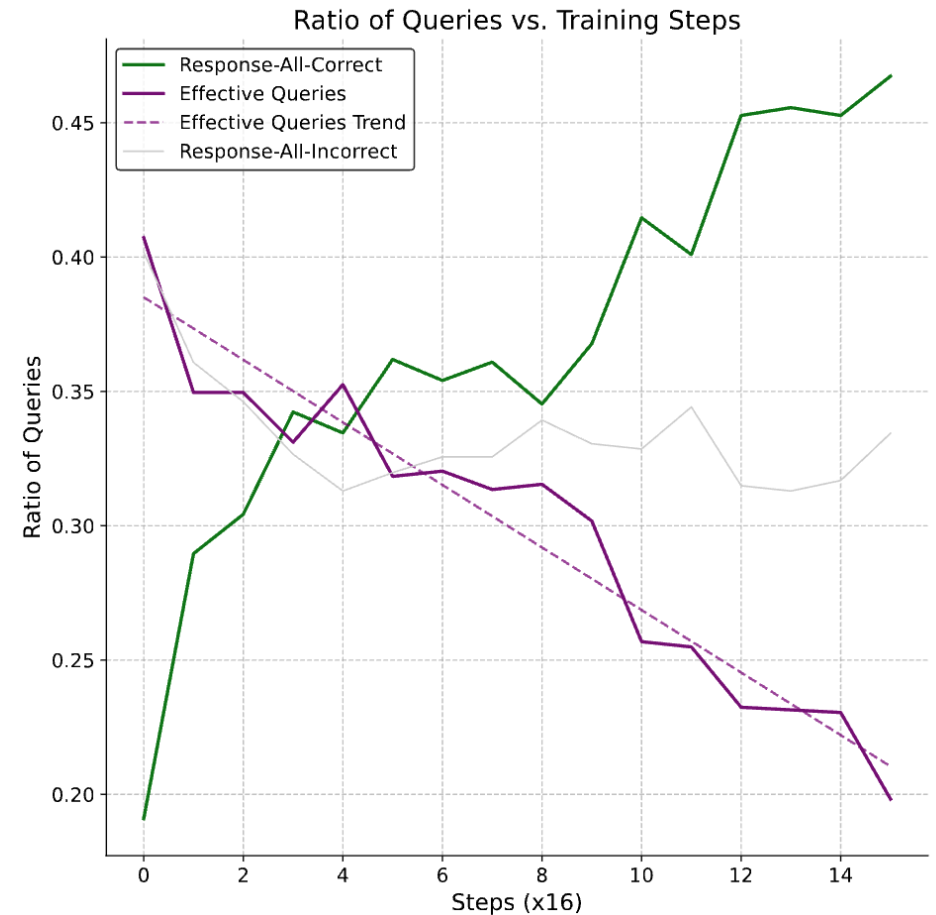
- Advantage becomes nearly zero when rewards are similar
 - Results in no or weak learning signal

Too hard
 $\{r_1, r_2, \dots, r_G\} = 0$

$\{r_1, r_2, \dots, r_G\} = 1$

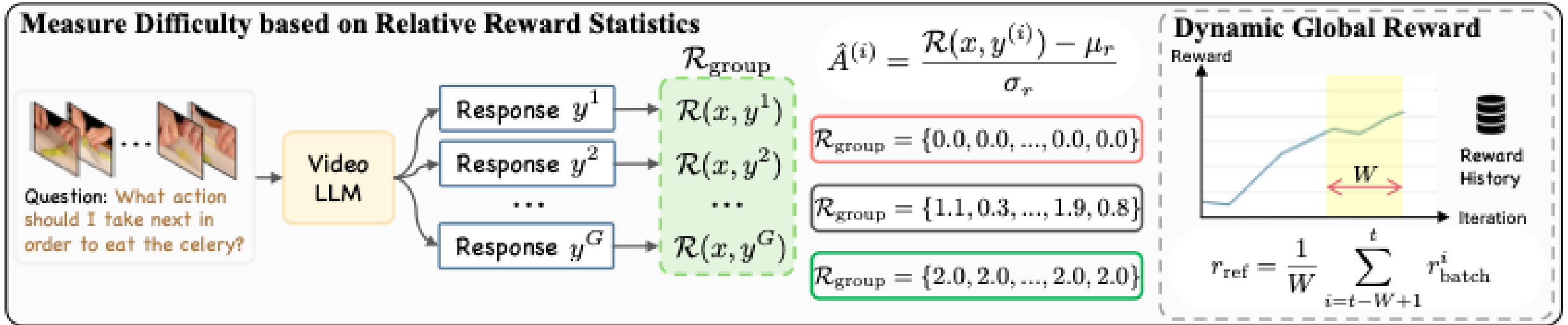
Too easy!

$$\hat{A}_{i,t} = \tilde{r}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$$



Method

Sol 2. Difficulty-aware Data Augmentation



- Define sample difficulty and adjust it to a moderate level

How is difficulty measured?

Compare the current group's mean reward with dynamic global mean (sliding window)

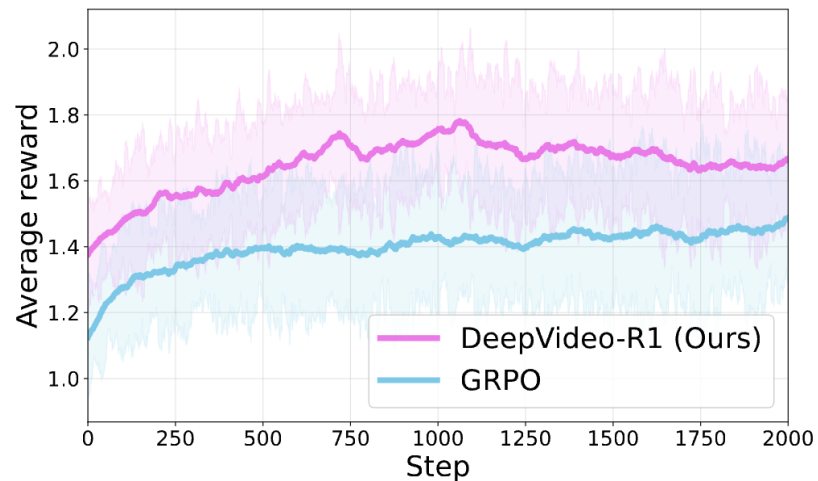
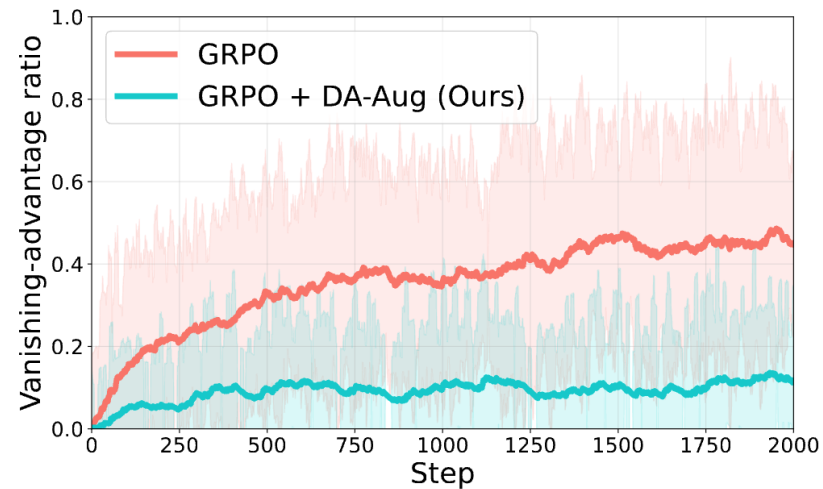
- Lower mean → **harder** sample
- Larger mean → **easier** sample

Experiment

Quantitative Results

Table 1: Performance on SEED-Bench-R1 validation split. In-Dist. means in-distribution dataset while OOD indicates out-of-distribution.

Method	SBR-L1 (In-Dist.)	SBR-L2 (OOD)	SBR-L3 (OOD)				Overall
	Epic-Kitchens Daily life	Ego4D Daily life	Daily life	Hobbies	Ego4D Recreation	Work	
Baseline							
VideoLLaMA3-7B [8]	33.33	33.21	26.65	28.47	30.60	26.98	27.71
InternVL3-2B [6]	23.71	23.14	21.18	16.27	18.58	12.62	17.11
InternVL3-8B [6]	41.41	40.75	39.18	34.58	34.97	29.95	34.75
InternVL3-14B [6]	43.61	45.79	43.96	34.92	36.61	31.68	37.17
Qwen2-VL-2B							
Qwen2-VL-2B [52]	12.88	16.35	12.98	16.27	10.38	14.36	13.78
Qwen2-VL-2B + SFT	34.05	36.23	36.90	34.58	29.95	33.88	33.84
Qwen2-VL-2B + GRPO	38.39	42.01	40.55	37.63	30.45	40.44	36.79
DeepVideo-R1-Qwen2-VL-2B (Ours)	48.88	50.31	52.39	42.71	41.09	49.18	46.33
Qwen2-VL-7B							
Qwen2-VL-7B [52]	34.79	34.02	31.21	32.30	33.33	30.69	31.57
Qwen2-VL-7B + SFT	43.79	44.10	38.27	41.02	32.24	38.61	38.15
Qwen2-VL-7B + GRPO	46.01	50.16	48.52	45.08	43.72	41.34	44.89
DeepVideo-R1-Qwen2-VL-7B (Ours)	56.39	59.75	57.63	52.54	50.00	55.19	53.82
Qwen2.5-VL-3B							
Qwen2.5-VL-3B [5]	31.32	32.70	33.03	28.81	27.32	23.02	28.24
Qwen2.5-VL-3B + SFT	35.90	39.12	39.86	31.53	29.70	31.15	33.69
Qwen2.5-VL-3B + GRPO	39.56	41.01	38.95	33.90	36.61	31.93	35.35
DeepVideo-R1-Qwen2.5-VL-3B (Ours)	48.05	51.07	46.47	45.76	43.72	40.10	43.98
Qwen2.5-VL-7B							
Qwen2.5-VL-7B [5]	33.41	38.24	35.08	31.53	27.32	27.97	31.04
Qwen2.5-VL-7B + SFT	42.37	42.64	41.23	37.29	36.63	41.53	38.99
Qwen2.5-VL-7B + GRPO	49.07	52.08	49.43	40.68	43.17	35.15	42.24
DeepVideo-R1-Qwen2.5-VL-7B (Ours)	51.95	55.72	51.25	47.80	46.99	44.06	47.69



Qualitative Results

Increased Reasoning

Question: Considering the progress shown in the video and my current observation shown in the image, what action should I take next in order to clean the house?

- A. collect dirt B. put down broom C. roll cable around vacuum cleaner
D. remove charger

Output the thinking process in <think> </think> and final answer in <answer> </answer> tags, i.e., <think> reasoning process here </think><answer> answer here </answer>.



Qualitative Results

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D. remove charger

Output the thinking process in <think> </think> and final answer in <answer> </answer> tags, i.e., <think> reasoning process here </think><answer> answer here </answer>.



GRPO: <think> The person in the video is holding a broom, which suggests that they are in the process of cleaning the house. The presence of the broom indicates that the next logical step would be **to continue using it to collect dirt and debris**. The other options do not seem relevant to the current activity shown in the video.</think> <answer>**A. collect dirt**</answer>

DeepVideo-R1(Ours): <think> The person in the video is currently cleaning the floor with a broom. The next logical step would be **to put down the broom after finishing cleaning the floor**.</think> <answer> **B. put down broom**</answer>

Qualitative Results

Increased Reasoning

Question: Considering the progress shown in the video and my current observation shown in the image, what action should I take next in order to prepare berries?

A. move tray B. close tray C. open container **D. move berries**

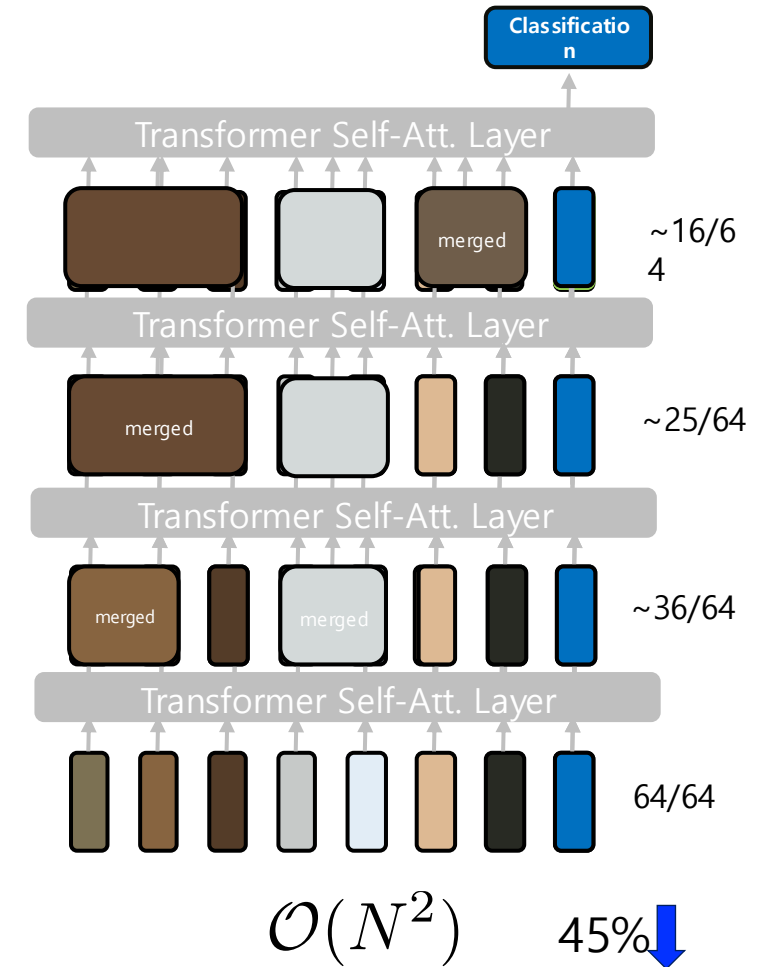
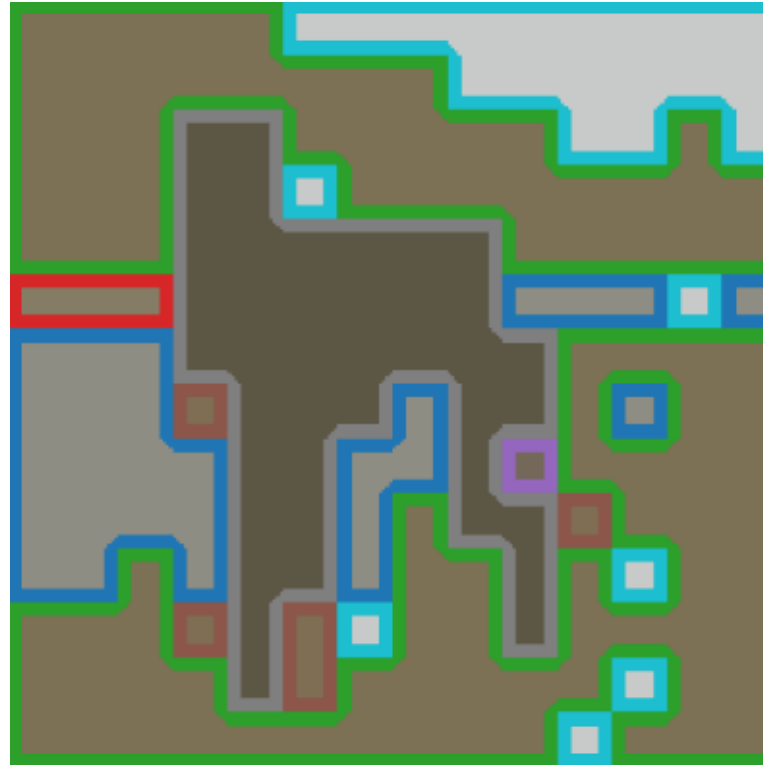
Output the thinking process in <think> </think> and final answer in <answer> </answer> tags, i.e., <think> reasoning process here </think><answer> answer here </answer>.



GRPO: <think> The image shows a person preparing food, specifically making a dish with vegetables and cheese on a flatbread. **There are no berries or trays visible in the image**, so it is not possible to determine the next steps for preparing berries. The options provided are related to trays and containers, which are not relevant to the current task of preparing berries. </think> <answer> **C. open container** </answer>

DeepVideo-R1(Ours): <think> The person in the video is currently preparing a dish with berries. The next logical step would be **to add more berries to the dish or to arrange the existing berries in a specific way**. Since the image shows the person already handling the berries, the next action should be to **add more berries or to arrange the existing ones**. </think> <answer> **D. move berries** </answer>

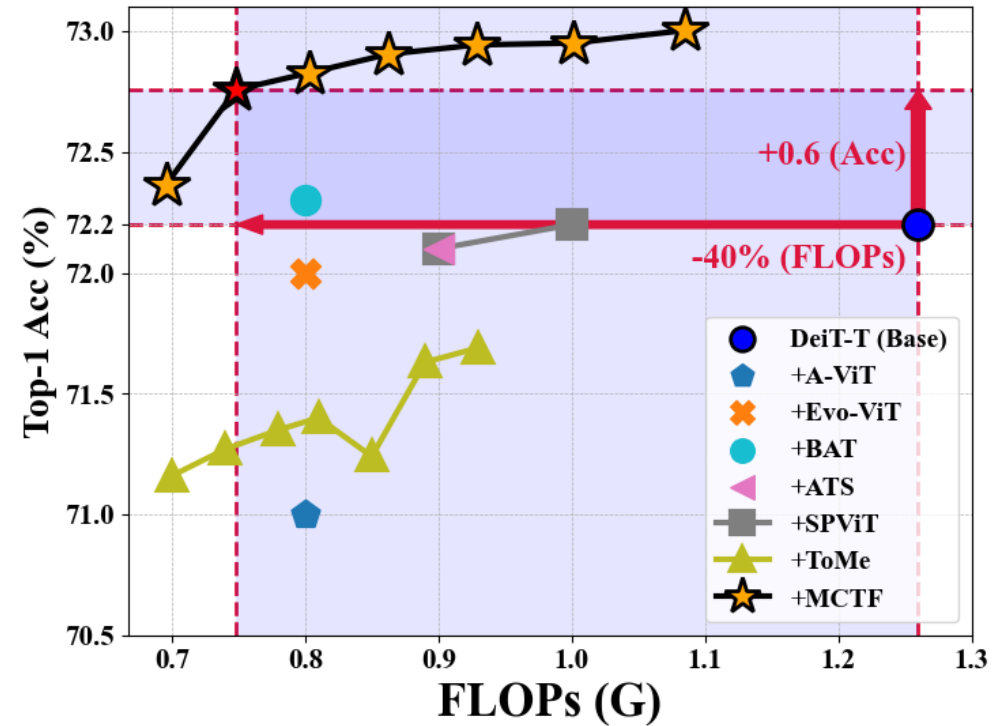
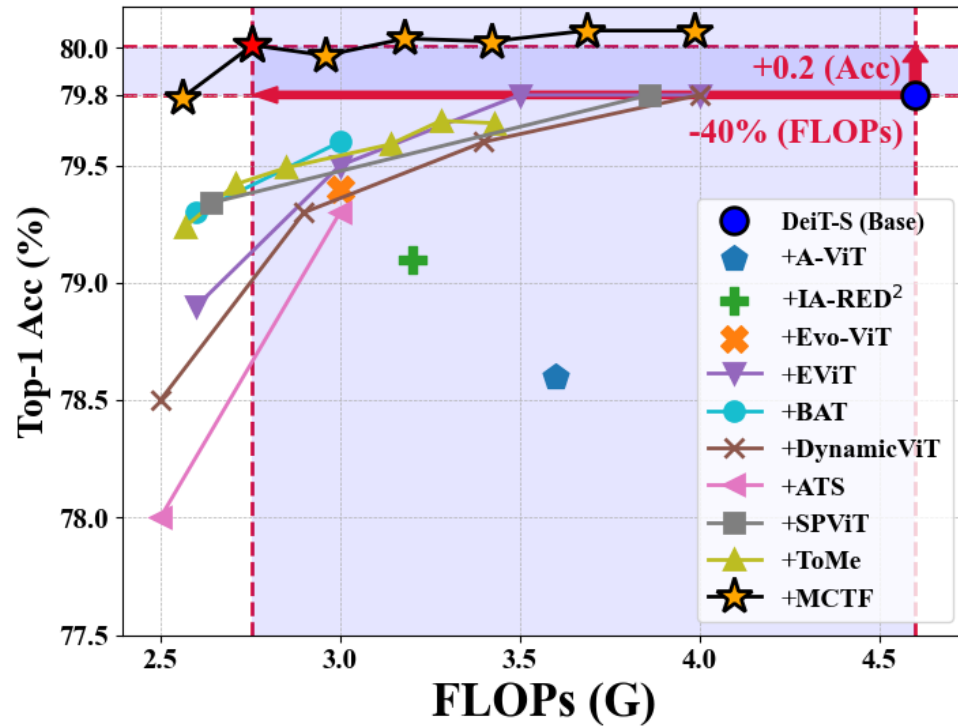
Token Mering



[Bolya et al., ICLR 2023] Token merging: Your vit but faster

*[Lee et al., CVPR 2024] Multi-criteria Token Fusion with One-step-ahead Attention for Efficient Vision

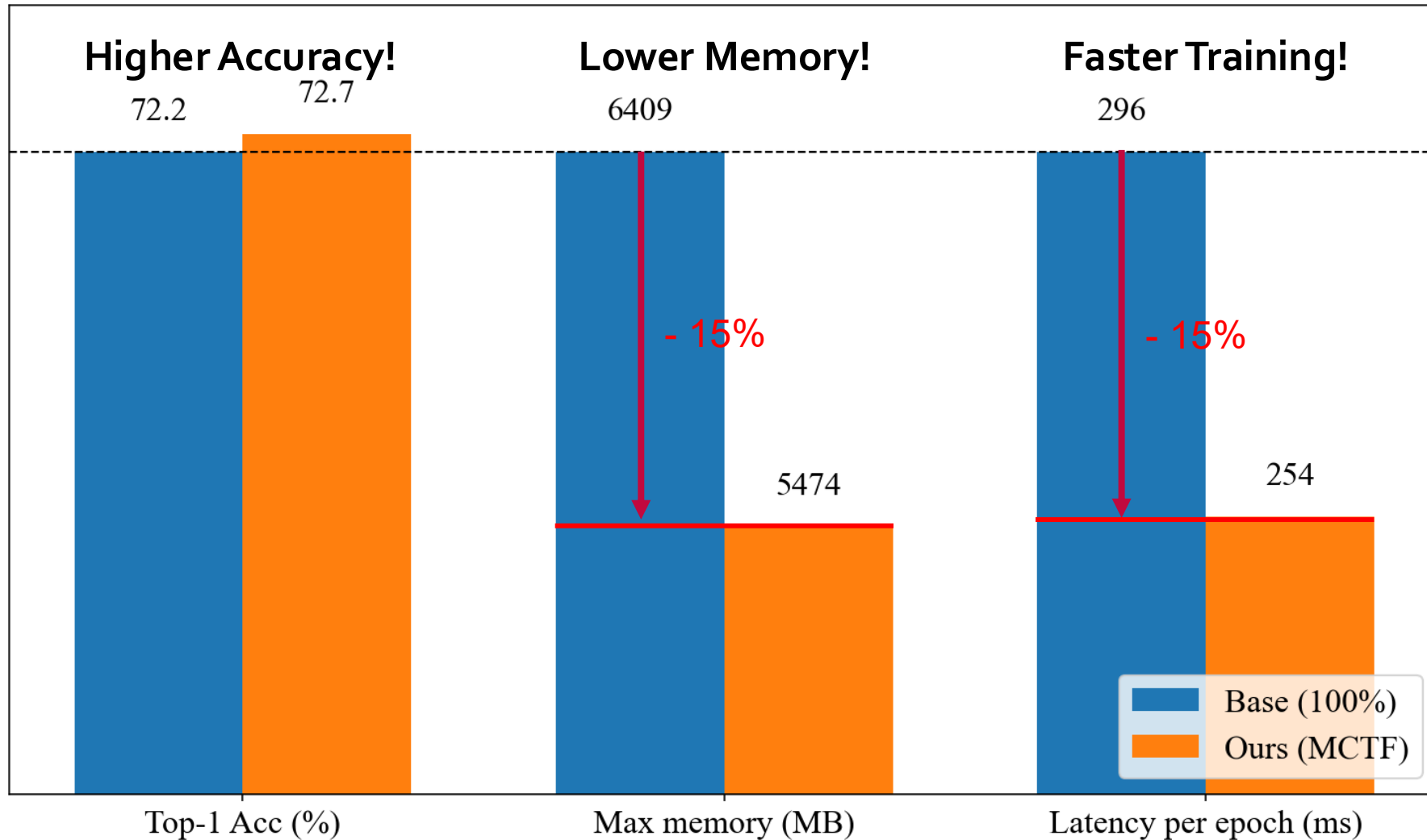
Multi-criteria Token Fusion



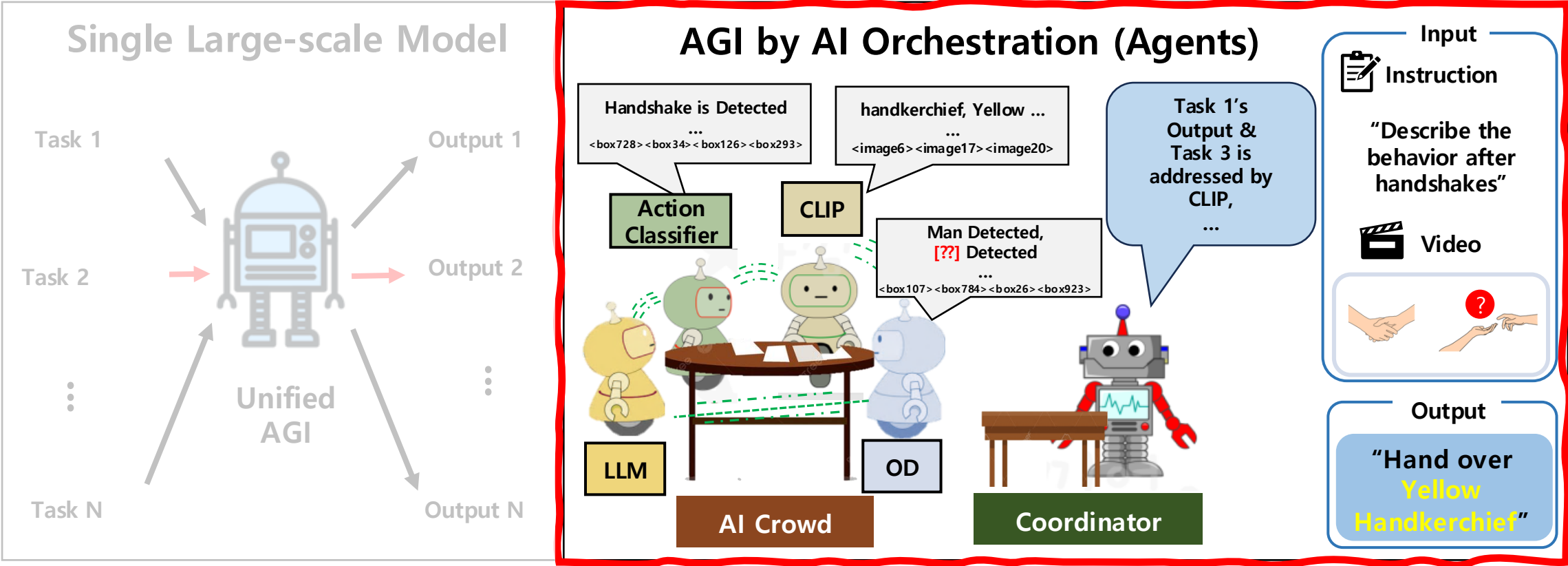
*[Lee et al., CVPR 2024] Multi-criteria Token Fusion with One-step-ahead Attention for Efficient Vision

*[Choi et al., CVPR24] vid-TLDR: Training Free Token merging for Light-weight Video Transformer

Speed-up training via Token merging



From a single model to Agents



2025 Will Be the Year of AI Agents

According to Sam Altman, **2025 will be the year AI agents go mainstream**. These systems won't just respond to commands—they'll act on their own, handling complex tasks without constant human input. This shift could transform industries like healthcare, science, and education. Imagine an AI assistant that not only answers your questions but proactively completes tasks you haven't even requested.

Altman predicts this leap is coming faster than expected. The o1 model is set to improve rapidly in the coming months, introducing AI with more advanced reasoning and task management capabilities. These aren't just chatbots anymore—AI agents will manage complex, multi-step processes autonomously.



In healthcare, for example, AI has already **sped up drug discovery by 10x**, and the market is expected to reach **\$6.6 billion by 2025**. With these advancements, AI could revolutionize diagnostics, patient care, and

<https://felloai.com/2024/10/sam-altman-reveals-openais-next-big-leap-you-not-gonna-believe-it/>

<https://www.youtube.com/watch?v=sal78ACtGTc&t=31s>



Andrew Ng
AI Fund

Dense Video Captioning

Dense Video Captioning (DVC)



Untrimmed Long Video

Goal:

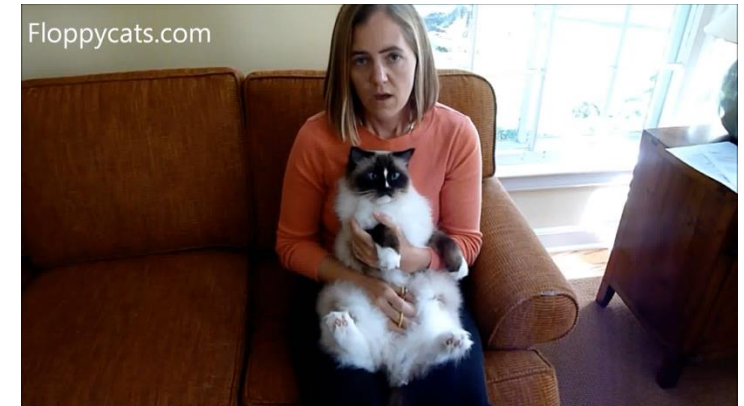
Generate a **concise video story** that supports human video comprehension without watching it.

Output:

A woman is seen holding on a cat on a couch and holding up a pair of nail clippers, *from 00 to 25*.
She begins cutting the cat's claws while the cat squirms around a bit, *from 25 to 72*.
She continues cutting the claws while speaking to the camera, *from 72 to 93*.



OpenAI ChatGPT 4.0



Could you outline the incidents that occurred at various timestamps in the video?



I can analyze the video for timestamps and provide an outline of incidents. Let me process the video to identify key events and generate a summary.

Key events
detection failed

The analysis did not detect any specific incidents based on the simplified approach used (e.g., brightness changes or motion thresholds) To provide a detailed outline of incidents, I could manually inspect the video or apply advanced processing techniques. Let me know how you'd like to proceed! [>-]

Fails to identify the semantically meaningful major events in the video.



Qualitative Results of VidChain



Could you please detail the events that took place during different time segments in the video?



Baseline



A woman is seen sitting on a couch with a cat in her lap, from 00 to 17.
She clips the cat's nail and then pets him, from 17 to 33.
She then clips the cat's other paw and pets him, from 33 to 50.
She clips the cat's other paw and pets him, from 67 to 83.
She clips the cat's other paw and pets him, from 83 to 99.

Too many segments
Too repetitive

Method: ① Chain-of-Tasks (CoTasks)

CoTasks

How many of time segments can this video breakdown into?

3 time segments.

Can you breakdown the video into different time segments?



00

25



25

72



72

99

Could you please detail the events that took place during different time segments in the video?

From 00 to 11, A woman is seen holding on a cat on a ...

From 12 to 55, She begins cutting the cat's claws while the...

From 55 to 99, She continues cutting the claws while speak..

Question 1:

How many of time segments can this video breakdown into?

Answer 1:

3 time segments.

Question 2:

Can you breakdown the video into different time segments?

Answer 2:

First event from 00 to 25.

Second event from 25 to 72.

Third event from 72 to 99.

Question 3 :

Could you please detail the events that took place during different time segments in the video?

Answer 3:

From 00 to 25, a woman is seen holding on a cat on a couch and ..

From 25 to 72, She begins cutting the cat's claws while the cat squirms..

From 55 to 99, She continues cutting the claws while speaking to the...

Illustration of the CoTasks approach

Method: ② Metric-based Direct Preference Optimization (M-DPO)

Provide
task-specific
supervision
during training

Task 1:

Number of event prediction
 $p(n|v)$ – **Accuracy**

Task 2:

Temporal video grounding
 $p(t|n, v)$ – **IoU**

Task 3:

Fine-grained video captioning
 $p(c|t, n, v)$ – **CIDEr, METEOR**

Question 1:

How many of time segments can this video breakdown into?

Answer 1:

3 time segments.

Question 2:

Can you breakdown the video into different time segments?

Answer 2:

First event from 00 to 25.

Second event from 25 to 72.

Third event from 72 to 99.

Question 3 :

Could you please detail the events that took place during different time segments in the video?

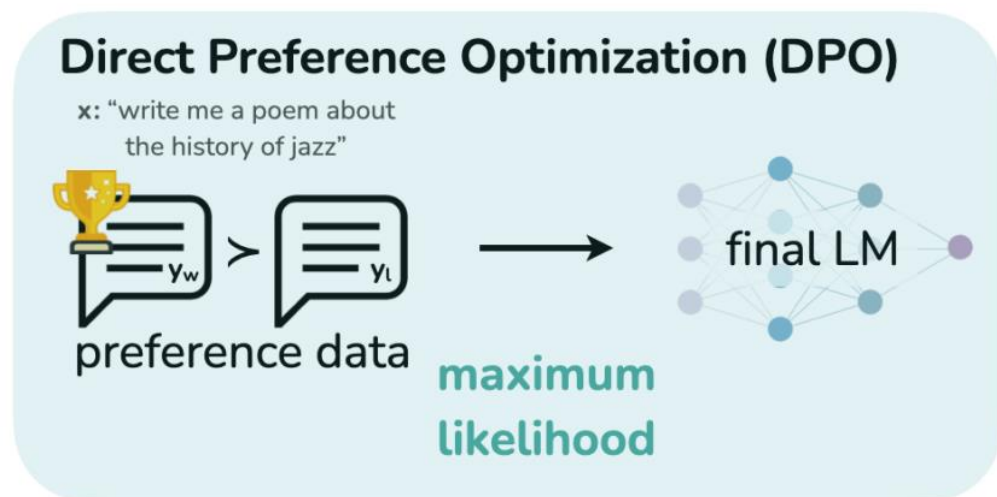
Answer 3:

From 00 to 25, a woman is seen holding on a cat on a coach and ..

From 25 to 72, She begins cutting the cat's claws while the cat squirms..

From 55 to 99, She continues cutting the claws while speaking to the...

Method: ②Metric-based Direct Preference Optimization (M-DPO)



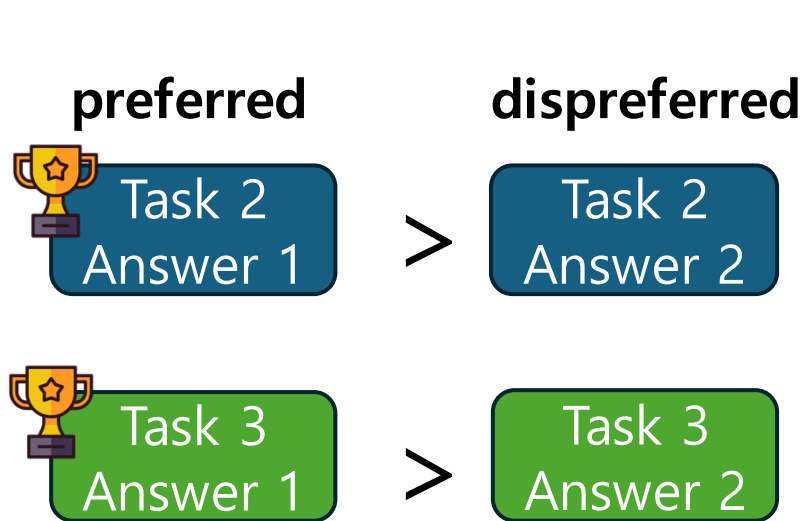
DPO allows to align Large Language Models' output with **human preferences**.

Train the model to increase the likelihood of the preferred response relative to that of dispreferred response

DPO Objective:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(v, x, \hat{y}^w, \hat{y}^l) \sim \mathcal{D}_{\text{DPO}}} \left[\log \sigma \left(\beta \cdot r(\hat{y}^{\text{win}}; v, x) - \beta \cdot r(\hat{y}^{\text{lose}}; v, x) \right) \right]$$
$$r(y; v, x) = \log \frac{\pi_{\theta}(y|v, x)}{\pi_{\text{ref}}(y|v, x)}$$

Method: ②Metric-based Direct Preference Optimization (M-DPO)



$(\mathcal{P}_{c \rightarrow t})$

Metric preference for $p(c|n, v)$:
one with higher **caption** quality (e.g., **METEOR**)

Metric preference of $p(t|c, n, v)$:
one with higher **IoU/SODA** score

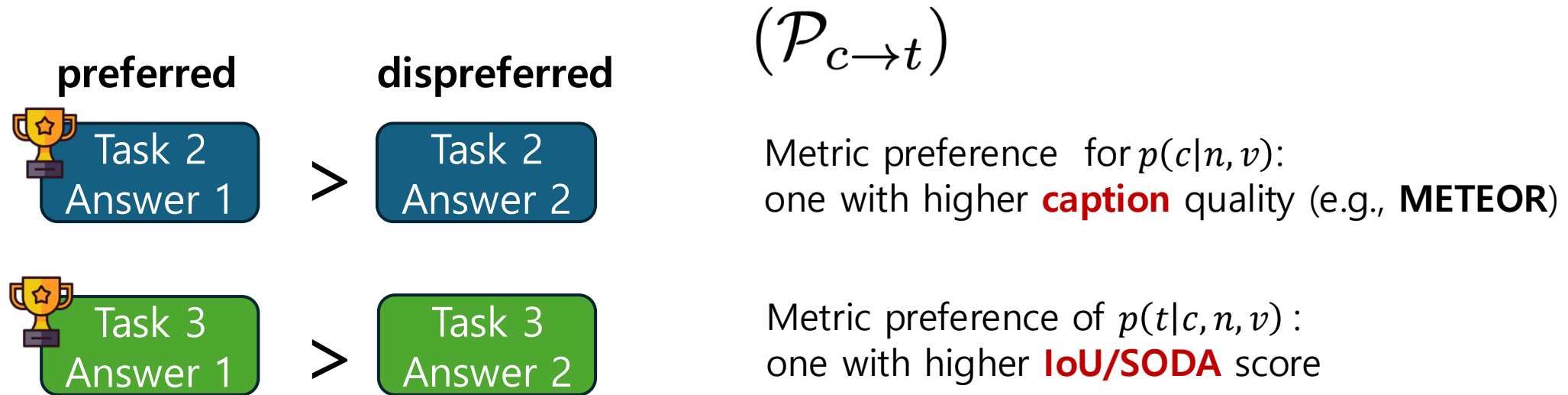
M-DPO Objective:

$$\mathcal{L}_{\text{M-DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{\mathcal{D}_{\text{M-DPO}}} \left[\mathbb{1} (m_k^w - m_k^l > \gamma) \cdot \mathcal{L}_s (\hat{y}_k^w, \hat{y}_k^l; v, h_{<k}, x_k) \right]$$

Intermediate supervision

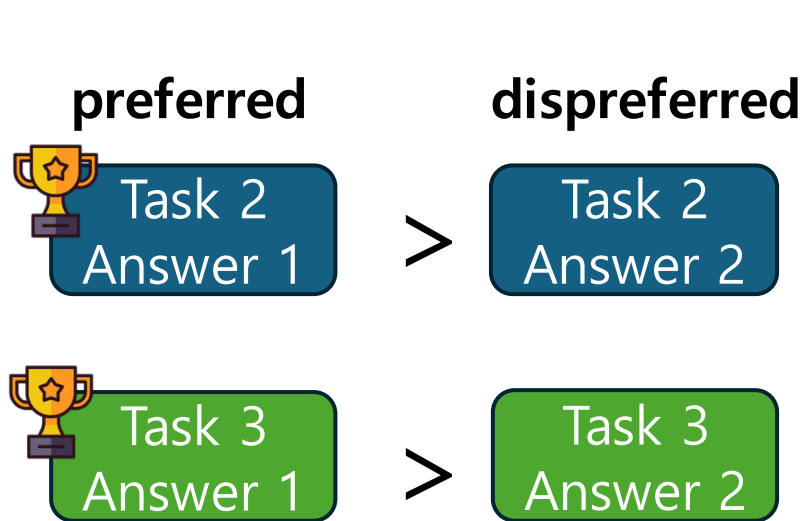
$$\mathcal{L}_s(\hat{y}_k^w, \hat{y}_k^l; v, h_{<k}, x_k) = [\log \sigma (\beta r(\hat{y}_k^w; v, h_{<k}, x_k) - \beta r(\hat{y}_k^l; v, h_{<k}, x_k))]$$

Method: ②Metric-based Direct Preference Optimization (M-DPO)



- **Metric preferences** reveals **how much** the output is preferred
“How much” – select samples where the preference gap is above a certain threshold γ

Method: ②Metric-based Direct Preference Optimization (M-DPO)



$(\mathcal{P}_{c \rightarrow t})$

Metric preference for $p(c|n, v)$:
one with higher **caption** quality (e.g., **METEOR**)

Metric preference of $p(t|c, n, v)$:
one with higher **IoU/SODA** score

M-DPO Objective:

$$\mathcal{L}_{\text{M-DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{\mathcal{D}_{\text{M-DPO}}} \left[\mathbb{1} \left(m_k^w - m_k^l > \gamma \right) \cdot \mathcal{L}_s \left(\hat{y}_k^w, \hat{y}_k^l; v, h_{<k}, x_k \right) \right]$$

Metric preference gap aware Intermediate supervision

$$\mathcal{L}_s(\hat{y}_k^w, \hat{y}_k^l; v, h_{<k}, x_k) = [\log \sigma(\beta r(\hat{y}_k^w; v, h_{<k}, x_k) - \beta r(\hat{y}_k^l; v, h_{<k}, x_k))]$$

Qualitative Results of VidChain



Could you please detail the events that took place during different time segments in the video?



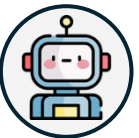
Baseline



A woman is seen sitting on a couch with a cat in her lap, from 00 to 17.
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She then clips the cat's other paw and pets him, from 33 to 50.
She clips the cat's other paw and pets him, from 67 to 83.
She clips the cat's other paw and pets him, from 83 to 99.

Too many segments
Too repetitive

Ours



A woman is seen sitting on a couch holding a pair of clippers and speaking to the camera, from 00 to 25.
The woman then cuts the nails of a cat in her lap while the cat moves its feet, from 25 to 74.
The woman continues cutting the nails and is shown speaking to the camera, from 73 to 99.

Concise & accurate
Depicts the major events

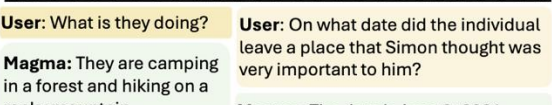
Vision Language Action (VLA) Models

Magma



User: What is they doing?

Magma: They are camping in a forest and hiking on a rocky mountain.



User: On what date did the individual leave a place that Simon thought was very important to him?

Magma: The date is June 9, 2021.

Temporal



User: The figure represents a 5x5 grid containing various animals, where each 1x1 square is considered a block and each block contains an animal from ['cat', 'dog', 'elephant', 'giraffe', 'rabbit']. What is the animal of the block located at the first row, second column of the grid?

Magma: The animal is a cat.

Spatial



User: What is unusual thing about this image?

Magma: The unusual aspect of this image is that a man is ironing clothes on the back of a yellow taxi.



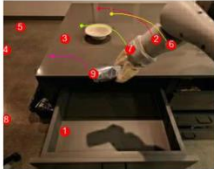
User: How to increase the sound?

Magma: Tap on the "Volume" icon. Then, slide the volume up to increase the volume.

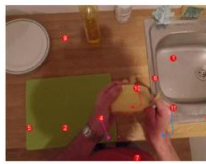
Multimodal Understanding



Task: Push the chip bag to left



Task: Put the can to left



Task: Peel the potato



Task: Stir the food in the pot

Planning



Task: Plan a visit to the museum



Task: Lock Screen



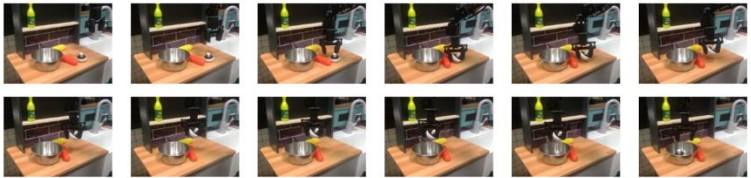
Task: View my Account

Multimodal Action Grounding



User: Close the middle drawer

Magma: Trajectory Rollout on SimplerEnv Google Robot



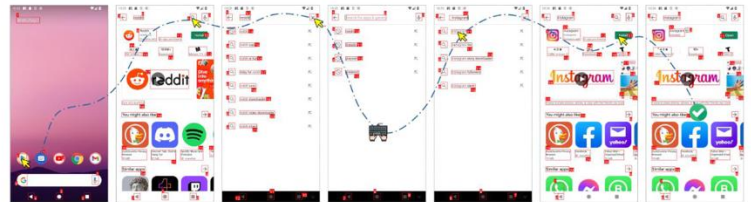
User: Pick up the mushroom to the pot

Magma: Trajectory Rollout on Widow-X Real Robot



User: Find baby shoes under 20 dollars and rated 5 stars.

Magma: Trajectory on Mind2Web

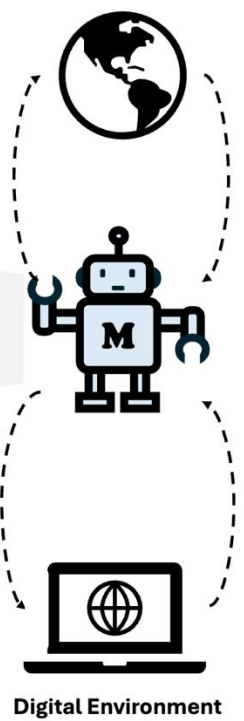


Task: Install app "Instagram"

Magma: Trajectory on AITW

Multimodal Agentic Taks

Physical Environment



Digital Environment

✓ A model capable of not only Multimodal Understanding, but also Action Grounding and Agentic Task

Magma (Core Challenges)

- ✓ **Input/output mismatch across diverse tasks during pre-training**
 - ✓ **UI navigation:** 2D screenshot input → predict 2D (x,y) coordinates
 - ✓ **Robot manipulation:** 3D video input → predict 3D position/pose
 - ✓ **Vision-language understanding:** image/video + text input → text output

- ✓ **Scarcity of action-focused robot data**
 - ✓ Unlike image-text data, robot datasets are very limited in volume — only on the order of a few million
 - ✓ As a result, securing sufficient action signal is difficult

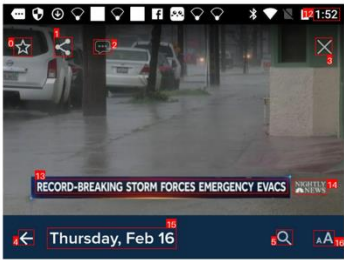
Vision Language Action (VLA) Models

Magma (Methodology)

- ✓ Define surrogate tasks
 - ✓ **Set-of-Mark (SoM):** assign numbered marks to actionable objects in the image so the model predicts the number instead of coordinates → reduced search space enables more efficient learning
 - ✓ **Trace-of-Mark (ToM):** predict the motion trajectories of objects/hands in video → enables temporal reasoning

✓

UI Navigation



Task: Swipe until Thursday, Feb 16.

Set-of-Mark:
Mark 15 for bounding box at coordinate:
[33, 95, 130, 105]

Robot Manipulation



Task: Move the white object above the green cloth.

Set-of-Mark:
Mark 1 at [169,54], Mark 4 at [152,37], Mark 8 at [169,86], Mark 9 at [169,70]

Human Action

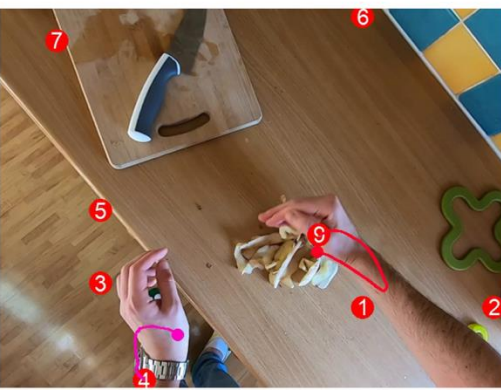


Task: Move potato in left hand to the center of cutting board.

Set-of-Mark:
Mark 1 at [72,231], Mark 4 at [56,243], Mark 6 at [105,182], Mark 7 at [121,117]

Set-of-Mark

Human Action



Task: Future 16 steps to move hands towards the pile of potato peels to gather them together.

Trace-of-Mark:
{
 "Mark 4":
 "[[85,248],[84,243],[84,237],[83,230],[83,224],[83,220],[84,217],[86,215],[89,214],[92,215],[96,216],[98,217],[99,219],[99,221],[98,222],[96,225]]",
 "Mark 9":
 "[[157,149],[160,149],[163,151],[166,155],[172,163],[175,173],[177,182],[178,185],[178,190],[176,189],[171,183],[164,174],[156,166],[149,162],[144,160]]"
}

Trace-of-Mark

Video Conversation



User: I really want to win this game. Can you suggest the next move that I should make to maximize my chances of winning?

Magma: I suggest moving the pawn to d4 to maximize your chances of winning.

UI Navigation

Agent in Digital World

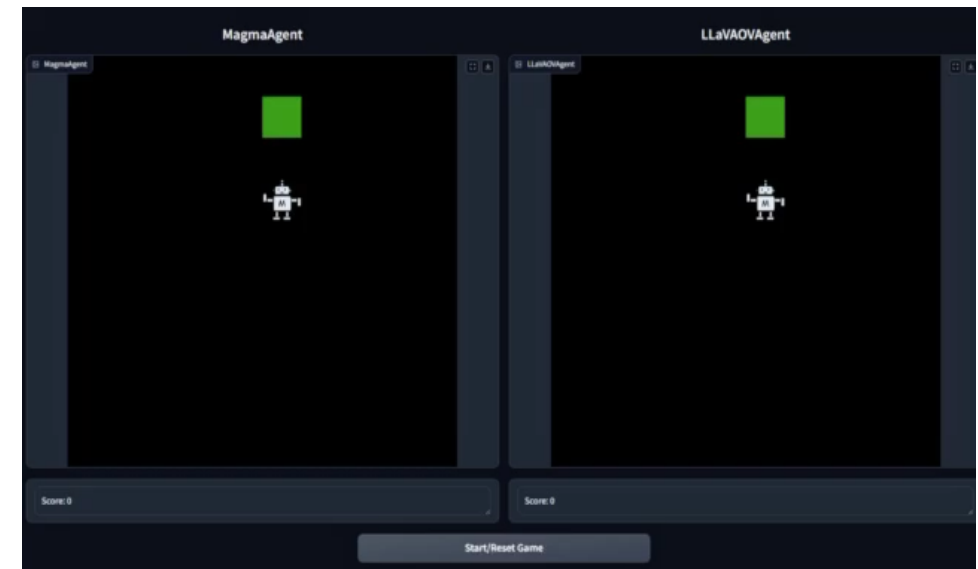
Robot Visual Planning



Robot Manipulation



Magma Gaming



Thank you!

